Can Educational Policies Reduce Wealth Inequality?

Fernando Loaiza

May 30, 2026

Abstract

This paper studies how education shapes wealth accumulation over the life cycle using U.S. panel data spanning two generations. Combining three empirical strategies with a quantitative life-cycle model, I find that college attainment generates a large and accelerating wealth premium: the education-wealth gap roughly doubles between individuals in their 30s and 60s. A channel decomposition shows that differential returns to capital, rather than income differences alone, are the main driver of this age-acceleration, accounting for nearly 60% of the empirical gradient. The model also identifies an income-risk mechanism: equalizing non-college income volatility to college levels changes saving behavior and alters the timing of wealth accumulation. Policy counterfactuals show that financial democratization, which equalizes asset returns across education groups, sharply compresses the gradient, whereas a modest expansion in college enrollment still leaves it nearly unchanged. Policies targeting financial market access may therefore reduce wealth inequality more effectively than enrollment expansion alone.

Keywords: Wealth Inequality · Returns to Education · Life Cycle · Heterogeneous Returns · Educational Policy

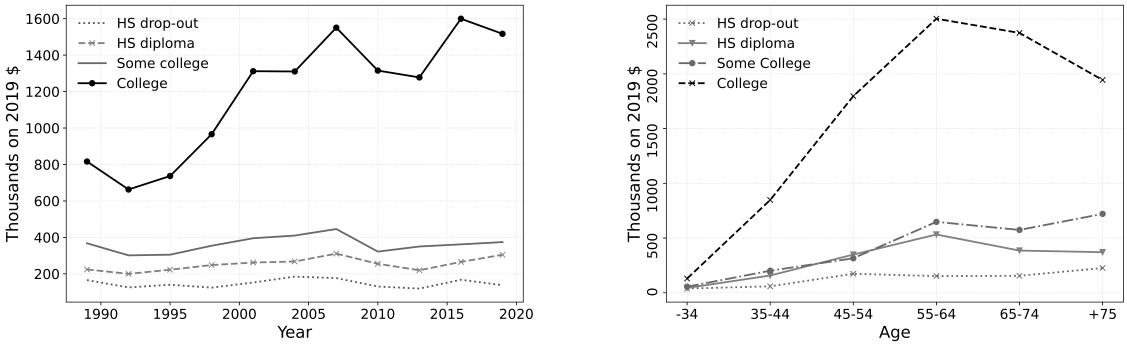
JEL Codes: D15 · D31 · E21 · I24 · I26

*Contact: Max Planck Institute for Innovation and Competition, Herzog-Max-Str. 4, 80333 Munich, Germany. Email: fernando.loaiza@ip.mpg.de. This paper was completed while the author was affiliated with the Max Planck Institute for Social Law and Social Policy.

1 Introduction

The distribution of wealth in the United States has grown markedly more concentrated over the past four decades. The top 1 percent of households now account for over 40 percent of total net worth, a level that stands out by both historical and international benchmarks (Alvaredo et al., 2018; Saez & Zucman, 2016). These facts have placed economic mobility near the top of the policy agenda, and education has long been the preferred answer (Mincer, 1958; Psacharopoulos & Patrinos, 2018). Federal and state governments devote substantial public resources to expanding college access, improving school quality, and subsidizing tuition, on the premise that human capital investment narrows economic disparities (Psacharopoulos & Patrinos, 2018; Heckman et al., 2006). The evidence on wages supports this intuition: a large body of research documents that an additional year of schooling raises labor earnings by approximately 8 to 10 percent (Card, 1999), and the college earnings premium has remained large and persistent for decades (Heckman et al., 2006). And yet wealth inequality has continued to rise precisely during the period when educational attainment expanded most rapidly. Between 1989 and 2019, the share of Americans with at least some college education grew substantially, while wealth concentration remained persistently high throughout (Alvaredo et al., 2018). Recent policy debates have centred on two distinct strategies: expanding who goes to college, through free tuition programs, Pell Grant expansion, and debt relief, and improving what college provides, through better curricula, teacher quality, and skill formation (Keller, 2010). These are not the same policies. This paper shows they have sharply different implications for wealth inequality, and that conflating them leads to misleading conclusions about whether education can close the wealth gap.

The difficulty in answering this question lies partly in the distinction between income and wealth as outcomes, and partly in an identification problem that is especially severe for wealth. Labor earnings respond to education year by year; wealth accumulates over decades through saving decisions, participation in financial markets (Campbell, 2006), investment returns, and intergenerational transfers (Charles & Hurst, 2003). These channels may respond to education very differently from wages do. Yet the literature on returns to education has focused almost exclusively on labor income. Direct evidence on education and wealth is sparse and largely confined to Scandinavia: Bingley and Martinello (2017) find no effect on retirement wealth in Denmark; Fagereng, Guiso, Holm,



(a) Education Level

(b) life-cycle Profile

Figure 1: Wealth by Education Level
Note: Panel (a) presents median net worth by education level for survey years 1989–2019. Panel (b) presents the cross-sectional wealth-age profile by education level in 2019; because respondents of different ages are observed at a single point in time, these profiles reflect both age and cohort effects and should not be interpreted as lifecycle trajectories for a single cohort. Source: Survey of Consumer Finances, 1989–2019.

and Pistaferri (2020) find no returns in Norway; Girshina (2019) finds a positive life-cycle association in Sweden; and Audoly et al. (2024) show, using Norwegian administrative tax records, that human capital is the primary driver of upward wealth mobility for those who rise through the middle of the distribution. Evidence for the United States, where wealth inequality is substantially more severe, is more limited: Barth et al. (2023) document that genetic endowments linked to educational attainment predict wealth through both earnings and financial returns, but their identifying variation is genetic rather than institutional, leaving the mechanisms difficult to disentangle from correlated unobservables. The identification challenge is formidable: college-educated households hold substantially more wealth than high school educated households, as shown in Figure 1, but they also disproportionately come from wealthy families (Karagiannaki, 2017).[1] Parental wealth shapes both who attends college and what children eventually inherit, making it difficult to separate the return to education from the return to being born into advantage, particularly when family background is measured through income rather than wealth (Girshina, 2019).

This paper addresses both gaps. Using three decades of Panel Study of Income Dynamics data spanning two generations of U.S. households, it asks two related questions: does education predict sustained wealth accumulation over the life cycle, and through which

[1]See also Blanden and Machin (2004), Chevalier et al. (2013), and Atkinson and Bourguignon (2014) on the role of family background in educational and economic outcomes.

mechanisms does that relationship widen with age? To answer the first question, the analysis employs three complementary empirical strategies. A random effects panel model serves as a descriptive benchmark. A within-sibling comparison removes family-level confounders shared by siblings raised in the same household (Black et al., 2015). An instrumental variables approach exploits state-level variation in compulsory schooling laws (Acemoglu & Angrist, 2000) and parental job loss during a child's high school years (Stevens & Schaller, 2011) as sources of exogenous variation in educational attainment. Throughout, family background is controlled using directly measured parental wealth rather than income, which is the more relevant transmission channel. To answer the second question, the empirical findings inform a calibrated life-cycle heterogeneous agent model that decomposes the education-wealth gap into income, returns, and income-risk channels and then evaluates policy counterfactuals.[2]

The central empirical finding is that the education-wealth gap accelerates over the life cycle: the estimated college premium is substantially larger in the 50s and 60s than in the 30s across all three empirical strategies, consistent with a process of cumulative compounding rather than a one-time earnings shift. The quantitative model reproduces this age gradient and shows that the dominant mechanism is differential returns to capital. In the decomposition, the returns channel alone accounts for nearly 60% of the empirical age-acceleration, whereas the income channel by itself does not generate the observed pattern. The model also identifies an income-risk mechanism: equalizing non-college income volatility to college levels reduces the age gradient materially, implying that precautionary saving and financial stability are part of the education-wealth link. These results suggest that tertiary education affects wealth not only through labor earnings but through a broader set of financial channels that cumulate over the life cycle.

The policy simulations reinforce this asymmetry. Financial democratization, which equalizes asset returns across education groups, sharply compresses and even reverses the age gradient, while a modest expansion in the college share leaves the gradient nearly unchanged. A partial returns intervention, interpreted as financial education or improved financial capability, moves the gradient much less and can even steepen it slightly at low intensity because of compounding dynamics. The implication is not simply that "more

[2] Idiosyncratic returns to capital and their role in generating wealth concentration are discussed in Ma et al. (2020), Benhabib et al. (2019), and Daminato and Pistaferri (2024), who use PSID data to show that a persistent unobserved factor drives both return heterogeneity and earnings differences simultaneously.

education" reduces wealth inequality, but that the distributional effects of education policy depend on which margin is targeted: access, financial returns, or income stability. Policies that broaden financial market access or narrow return differentials (Lusardi & Mitchell, 2023; Kim, 2024) appear quantitatively more powerful than marginal enrollment expansion for compressing the education-wealth gap.

These findings contribute to three strands of literature. They provide comprehensive U.S. panel evidence on education and wealth accumulation, complementing and extending the Scandinavian evidence of Girshina (2019), Bingley and Martinello (2017), and Audoly et al. (2024) to a context with greater inequality and a larger college premium, and advancing on the genetic variation approach of Barth et al. (2023) by combining plausibly exogenous institutional instruments with a structural life-cycle decomposition. They connect the empirical returns to education literature Card (1999); Heckman et al. (2006) to the wealth inequality literature Ma et al. (2020); Benhabib et al. (2019); Daminato and Pistaferri (2024) by identifying the mechanisms through which education shapes wealth over the life cycle rather than wages alone. And they contribute to the literature on educational policy and redistribution Keller (2010); Lusardi and Mitchell (2023) by showing that the quantitatively more powerful policy margin is not merely college attendance, but the set of financial conditions under which wealth accumulates after schooling decisions are made Kim (2024).

The remainder of the paper is organized as follows. Section 2 presents the empirical analysis and discusses the identifying assumptions and limitations of each strategy. Section 3 develops and calibrates the quantitative model and presents the policy simulations. Section 4 discusses policy implications and limitations. Section 5 concludes.

2 Empirical Analysis

2.1 Empirical Model

This section employs three complementary empirical strategies to examine the relationship between education and wealth. Each strategy addresses different potential sources of confounding, but none constitutes a perfect natural experiment. Together, they provide evidence of a positive relationship between education, particularly tertiary education, and wealth accumulation. The three strategies are described in turn: a random effects (RE) panel model serving as a descriptive benchmark, a within-sibling comparison that reduces confounding from shared family background, and an instrumental variables approach that leverages sources of exogenous variation but rests on strong identifying assumptions. The RE model allows for unobserved individual-specific factors that are constant over time while retaining time-invariant variables of interest, mitigating bias from stable omitted variables, and thereby improving the precision of the estimated association between education and wealth. The baseline specification is given by:

$$\mathrm{Y}_{it} = \beta_0 + \beta_1\, \mathrm{Education}_i + \beta_2\, \mathrm{X}_i + \beta_3\, \mathrm{D}_{it} + \gamma_t + c_i + \upsilon_{it}, \tag{1}$$

where the indices i and t represent individuals and time, respectively. The dependent variable Y_{it} is total wealth, which varies over time. The key explanatory variables include both time-invariant and time-varying covariates. The time-invariant variables are $\mathrm{Education}_i$, measured as the individual's completed years of schooling, and X_i, a set of background characteristics recorded at age 16, including parental wealth, parental education, and proxies for individual ability; these variables do not change over the life cycle and therefore remain constant across all survey waves. Additional time-invariant factors are race and sex. In contrast, the time-varying controls D_{it} include demographic characteristics that evolve over time, such as age. The model also includes year fixed effects γ_t, which capture aggregate shocks common to all individuals in a given wave. Finally, c_i represents an unobserved, time-invariant individual-specific component, and υ_{it} is the idiosyncratic error term. The random effects specification is appropriate in this context because core explanatory variables, particularly education and family-background measures, are time-invariant and would otherwise be absorbed by a fixed effects estimator.

2.1.1 Reducing Family Confounding: Within-Sibling Variation

To reduce confounding from shared family background, a within-sibling variation (WS) strategy is implemented. It compares the wealth outcomes of two biological siblings who have made different schooling decisions. By differencing out shared environmental and genetic factors, this approach controls for common family-level unobservables such as parental wealth, parental education, and household norms. However, it does not achieve full causal identification: unobserved factors specific to one sibling, such as differential parental investment, birth-order effects, or idiosyncratic ability differences, may still confound the education-wealth relationship. This strategy is formalized as:

$$\Delta \mathrm{Y}_{jt} = \alpha_0 + \alpha_1\, \Delta \mathrm{Education}_j + \alpha_2\, \Delta \mathrm{X}_{jt} + \gamma_t + \upsilon_{jt}, \tag{2}$$

Here, ΔY_{jt} represents the wealth difference between siblings in pair j at time t. Since educational attainment is completed before observation begins and does not vary over time, the education difference $\Delta \mathrm{Education}_j$ carries only a sibling-pair subscript. The vector ΔX_{jt} contains within-pair differences in both time-invariant background characteristics, including socioeconomic conditions, parental presence during upbringing, participation in gifted programs, and class repetition, and time-varying controls such as age. γ_t accounts for time-fixed effects, and υ_{jt} is the error term. Despite the shared upbringing and genetic similarities, it is recognized that unobserved factors, such as differential parental support or knowledge transfer between siblings, could still influence education choices and net worth.

2.1.2 Instrumental Variables

While controlling for unobserved heterogeneity in parental background and individual abilities is important, it may not capture all pre-educational differences. To complement the previous strategies, a third approach leverages information on compulsory schooling laws (CSL) and parental job loss (PJL) for an instrumental variables analysis (IV). Under strong identifying assumptions, namely relevance, exogeneity, and exclusion, these instruments can help isolate variation in educational attainment that is plausibly unrelated to unobserved determinants of wealth. The results should be interpreted with caution, as each instrument's exclusion restriction relies on assumptions that cannot be directly tested and are discussed in detail below.

The first instrument utilizes the minimum required schooling years, matched to individuals based on the laws in their state when they were 14 years old. Since these laws vary by state and are considered exogenous, they serve as the basis for an instrumental variables approach. The analysis employs a two-stage least squares method, with the first stage predicting schooling based on compulsory education laws, while controlling for parental wealth to account for possible violations of the exclusion restriction:

$$\text{Education}_i = \beta_1 \, \text{CSL}_{s(i)} + \beta_2 \, \text{Par.Wealth}_i + \delta_s + \gamma_c + \epsilon_i, \tag{3}$$

where $\text{CSL}_{s(i)}$ denotes the compulsory schooling law requirement in state s matched to individual i based on their state of residence when they were 14 years old; δ_s are state fixed effects; and γ_c are cohort fixed effects. The variation in CSL is thus at the state level, not the individual level. Because Education_i is a completed, time-invariant measure, both the dependent variable and the error term carry only an individual subscript. The predicted values $\widehat{\text{Education}}_i$ from this first stage are then included in the second stage to examine the association between schooling and wealth. I specify the second stage as follows:

$$\text{Y}_{it} = \alpha_0 + \alpha_1 \, \widehat{\text{Education}}_i + \alpha_2 \, \text{Par.Wealth}_i + \delta_t + \gamma_c + \upsilon_{it}, \tag{4}$$

Here, Y_{it} indicates an individual's net worth. It also includes year fixed effects δ_t and cohort fixed effects γ_c. This approach assumes that compulsory schooling laws, as external factors, indeed affect educational attainment, a premise supported by Lochner (2010), who find that these laws are associated with substantial increases in educational attainment. The relevance of compulsory schooling laws as instruments for education, separate from direct effects on wealth, is further supported by findings in Acemoglu and Angrist (2000), highlighting their role in identifying the relationship between education and economic outcomes. Incorporating parental wealth into the analysis addresses potential concerns regarding the exclusion restriction, mitigating bias from shifts in compulsory schooling, potentially delaying young individuals' entry into the labor market, and prolonging financial dependence on parents.

While compulsory schooling laws provide valuable exogenous variation in educational attainment, some suggest they primarily affect high school attendance. They may not fully capture the impacts of tertiary education on wealth accumulation. Thus, a new instrument is introduced to address this limitation and complement the initial results: parental job loss during the child's final high school years. Parental job loss during this critical period introduces an exogenous shock to the family's financial stability, directly impacting the decision to pursue higher education. This event provides a source of exogenous variation that is particularly relevant for studying the effects of college education.

To support the validity of this instrument, parental wealth is controlled in both stages. Importantly, parental wealth is measured when the child is approximately 15 years old, that is, before or around the time of job loss, and thus captures the pre-shock level of family resources. While this pre-shock measure is a useful baseline control, it does not fully address all concerns about the exclusion restriction. Specifically, parental job loss may affect not only the child's college enrollment decision (the intended channel) but also future parental wealth and intergenerational transfers that occur after age 15. These post-shock transfers are not captured by wealth measured at age 15. As a result, the exclusion restriction may be partially violated if job loss reduces parental wealth at the time of the child's wealth accumulation, for instance, through lower bequests, inter vivos transfers, or informal financial support during adulthood. This concern is explicitly acknowledged, and the PJL instrument results should be interpreted as suggestive evidence, with the caveat that job loss may affect wealth outcomes through channels beyond the educational pathway. The first stage is presented by:

$$\text{College}_i = \beta_1\, \text{PJL}_i + \beta_2\, \text{Par.Wealth}_i + \gamma_c + \epsilon_i, \tag{5}$$

I specify the second stage as follows:

$$\text{Y}_{it} = \alpha_0 + \alpha_1\, \widehat{\text{College}}_i + \alpha_2\, \text{Par.Wealth}_i + \delta_t + \gamma_c + \upsilon_{it}, \tag{6}$$

where $\widehat{\text{College}}_i$ is the predicted probability of college attainment from the first stage, δ_t are year fixed effects, and γ_c are cohort fixed effects. As in the CSL strategy, the endogenous variable College_i is time-invariant and enters the second stage interacted with the panel structure through wealth Y_{it}, which varies over time.

The use of parental job loss as a complementary instrument offers a novel contribution to the literature on the effects of education and wealth. While compulsory schooling laws primarily influence high school completion, previous studies have shown that these reforms also have spillover effects on postsecondary attainment by shifting educational expectations, delaying labor-market entry, and leading to spillover effects on college enrollment and not just high school completion (Lochner (2010); Lavecchia et al. (2016)). This mechanism broadens their relevance for studying long-term wealth accumulation, as individuals exposed to stricter schooling laws are more likely to pursue further education.

Conversely, parental job loss operates at the upper educational margin, directly affecting the transition from high school to college by tightening family liquidity constraints precisely when higher-education decisions are made (Stevens and Schaller (2011); Charles et al. (2018)). Hence, the two instruments capture complementary dimensions of exogenous variation, policy-induced and financially driven, jointly addressing endogeneity while identifying both the extensive and intensive margins of education's impact on wealth.

2.2 Data and Sample Selection

This study utilizes data from 1999 to 2019 from the Panel Study of Income Dynamics (PSID), which captures the socioeconomic variables of families and their descendants over time, including comprehensive household financial wealth data from the wealth module initiated in 1984. The analysis employs two distinct samples to investigate parent-child and sibling relationships, focusing on individuals aged 30 or older who were heads of their family units (FUs). For intra-generational comparisons, the sample is limited to men due to higher data availability. It is assumed, for both samples, that by age 30, individuals have completed their education and begun accumulating wealth, consistently reporting the same level of education across different survey periods. All analyses use the Core–Immigrant family weights provided in each PSID wave.

Both samples exclusively consider biological relationships to minimize unobserved heterogeneity. Household wealth is total net worth excluding home equity. Because wealth data often include zero and negative values and exhibit substantial right-skewness, I apply a generalized inverse hyperbolic sine (IHS) transformation. This transformation accommodates non-positive values while mitigating the influence of extreme outliers, providing a log-like interpretation without requiring strict positivity. A scaling parameter is used following best practice in the literature to enhance numerical stability, given the wide range of monetary values observed.[3]

Education is operationalized differently across the three strategies to match each design's identifying variation.[4] In the RE and within-sibling models, education is entered as a set of five mutually exclusive categories based on completed years of schooling: high school dropout (0–11 years), high school diploma (12 years), some college (13–14 years), college

[3]Details and formal definitions of the IHS transformation are provided in Appendix A.2.

[4]Table A2 reports the full classification used in the categorical specifications.

degree (15–16 years), and postgraduate (17 or more years). In the CSL instrument, the endogenous variable is a binary indicator for *tertiary education* (college degree or above, i.e. 15 or more years of schooling), which is more plausibly affected by compulsory schooling laws through spillover effects on college enrollment. The same binary definition is used for the PJL instrument, where parental job loss is most directly linked to the college-going decision at the end of high school. The term "tertiary education" used throughout refers to this college-or-above category. Additional controls include parental wealth and education, captured from the earliest available PSID waves to approximate conditions when the Focus Unit was young, leveraging the survey's intergenerational design to account for family background. A family-level IQ measure serves as a proxy for ability, and I further control for age, sex, race, family structure by age 16, and inheritance receipt.[5]

The data on parental job loss is obtained from the PSID, exploiting its inter-generational features. The PSID provides detailed information on the employment status of household heads, including the number of weeks unemployed each year. We calculate parental job loss by summing the hours of unemployment for each parent during the years when the child is between 15 and 18 years of age. This period is chosen because it represents the critical high school years, during which financial stability can significantly influence a child's decision to pursue higher education. The data on compulsory schooling laws is sourced from Acemoglu and Angrist (2000). These laws provide an exogenous variation in educational attainment, essential for the instrumental variable approach, summarized as the higher of the minimum schooling years required or the difference between dropout and enrollment age requirements.

2.3 Empirical Results

Table 1 presents the descriptive benchmark results from the random effects regression, documenting the association between education and wealth across the life cycle. This analysis, which should be read as a descriptive characterization rather than a causal estimate, explores patterns by educational category. Education is a strong predictor of wealth even after controlling for background characteristics, highlighting persistent

[5]Summary statistics are presented in Table A3 and a descriptive analysis of wealth by education and cohort in Table A4, all in the Appendix.

differences in wealth accumulation across educational groups. The results highlight a progressive increase in wealth with higher education levels. For instance, individuals with a high school diploma see a wealth increment, which is significantly amplified for those with one to two years of college education. This trend continues, with postgraduate education showcasing the most substantial wealth gains, especially pronounced in the later life stages. For the remaining variables, inheritance and parental wealth consistently contribute positively across all models and life stages. Interestingly, the coefficients for parental education effects vary, showing a more positive and significant effect for fathers than mothers.

Table 1: RE Regression: Effects of Education on Wealth

	Avg	**Cohort**			
		30	**40**	**50**	**60**
High school	1220.52*	1832.99**	4196.66***	5523.55***	2089.51
	(611.31)	(629.30)	(649.82)	(867.19)	(1393.96)
Some College	2429.58***	1903.49**	5569.92***	6265.32***	8461.72***
	(677.35)	(697.32)	(768.53)	(966.68)	(1564.55)
College	2439.55**	5044.87***	10598.11***	10108.45***	10385.52***
	(783.29)	(791.38)	(866.67)	(1089.23)	(1685.97)
Postgraduate	2606.89**	1007.09	7252.09***	13484.91***	17702.61***
	(988.58)	(1135.31)	(1132.53)	(1258.68)	(1697.00)
Inheritance	0.15***	0.76***	0.51***	0.50***	0.53***
	(0.02)	(0.08)	(0.06)	(0.05)	(0.07)
Parental Wealth	0.28***	0.24***	0.22***	0.21***	0.16***
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)
Par.Education W.	362.50	258.52	641.68**	−175.27	−149.77
	(254.99)	(220.33)	(247.48)	(316.91)	(435.19)
Par.Education H.	597.69*	−13.73	807.43***	1463.31***	1206.75**
	(266.59)	(209.87)	(232.48)	(274.07)	(399.40)
Observations	20558	7028	6436	4825	1920
Adjusted R^2	0.26	0.16	0.23	0.28	0.37

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are heteroskedastic robust. The data uses sampling weights. Year, socio-demographic, and cohort effects are included. Socio-demographic variables include age, sex, and race. The constant term is included but not reported for brevity.

The within-sibling strategy, reported in Table 2, goes further by comparing siblings within the same family, thereby netting out shared family-level confounders. The results, for the average life cycle and by categories of education, are reported in the first column of the table, showing a consistently positive and significant association between education and wealth across almost all categories except for postgraduate education. While these estimates reduce family-level confounding, they should not be interpreted as causal, as within-family differences in unobserved individual characteristics may still drive part of the education-wealth relationship. A detailed examination of life stage-specific impacts reveals that education's positive influence on wealth persists across all age groups. Higher education levels correlate with increased wealth accumulation throughout the life cycle.

Table 2: Within Variation Regression: Effects of Education on Wealth

	Avg	**Cohort**			
		30	**40**	**50**	**60**
D.High school	1013.57^{+}	305.65	830.13	446.05	31983.15*
	(534.54)	(650.97)	(655.77)	(989.28)	(14720.33)
D.Some College	2605.85***	1671.88*	1729.37*	2948.22*	20230.82
	(616.07)	(706.41)	(853.63)	(1281.98)	(17346.99)
D.College	4985.44***	604.52	6665.88***	9253.42***	14464.63
	(986.05)	(1100.50)	(1369.20)	(2412.12)	(22087.41)
D.Postgraduate	747.67	−2691.06*	4075.86*	3204.44	40549.35
	(1172.68)	(1305.32)	(1669.10)	(3478.00)	(32783.38)
Observations	7887	3796	3078	1279	30
Adjusted R^2	0.02	0.04	0.05	0.07	0.00

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Time, socio-demographic, and cohort effects are included but not reported for brevity. Control variables include the difference between siblings in age, socioeconomic conditions, parental presence when young, and school performance. The constant term is included but not reported for brevity.

College education shows substantial wealth gains, particularly for individuals in their 40s and 50s, while the lack of significance in the 30 age group may be attributed to a delayed entry into the labor market due to the time spent in education. Postgraduate education shows positive wealth associations primarily for individuals in their 40s, but not consistently across other age groups. The negative sign in the first age group may reflect the opportunity cost of extended education. These patterns are also likely shaped by differences in sample size across groups: the 30s group contains 3,796 sibling pairs, the 40s 3,078, the 50s 1,279, and the 60s only 30 pairs, making the older-cohort estimates particularly noisy. The category 'Some College' education also exhibits significant effects, though with varying magnitudes. While these findings align with the previous strategy in indicating education's role in wealth accumulation, the within-sibling variation method generates new insights. Specifically, it reveals that the effects of education on wealth are most pronounced for tertiary education.

The instrumental variables strategy, introduced to complement the previous analyses, attempts to isolate exogenous variation in educational attainment. The first instrument leverages variation across U.S. states in compulsory schooling laws to examine how mandated education minimums are associated with long-term wealth accumulation. An initial regression is provided in Table A5 in the Appendix. The IV results are broadly consistent with the previous strategies, showing a positive relationship between education and wealth for the average and throughout the life cycle, though under the strong assumption that compulsory schooling laws affect wealth only through their effect on educational attainment. However, when separating education from a continuous variable into categories, it was found that lower levels of education do not have significant effects, nor for the average, nor for the different life-cycle stages. The only statistically significant effects were found for college and postgraduate education. These results suggest the notion that higher educational attainment, particularly at the college and postgraduate levels, plays a significant role in wealth accumulation throughout the life cycle, although with varied significance across different stages.

Table 3: I.V. Regression: Compulsory Schooling Laws on Tertiary Education

	Avg	**Cohort**			
		30	**40**	**50**	**60**
Tertiary	29958.92**	20842.69+	28197.41***	38114.43***	67838.94**
	(11405.44)	(10863.86)	(5829.78)	(7650.73)	(24456.19)
		First Stage			
CSL → College	0.02***	0.01***	0.02***	0.01***	0.01**
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
F-statistic	35.83	15.28	49.34	47.66	11.09
Observations	10281	1389	3912	3681	1243

Note: Source: Panel Study of Income Dynamics. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The instrument is the years of compulsory schooling by state. Year and cohorts effects are included. Parental wealth is included but not reported for brevity.

To address concerns regarding the validity of using compulsory schooling laws as an instrument for higher education, I created a binary variable distinguishing between tertiary education and non-tertiary education. By focusing on the first stage, this analysis examines whether CSL has a meaningful impact on higher education attainment, thereby providing evidence on instrument relevance and supporting the robustness of the IV strategy. The results, presented in Table 3, show a positive association between CSL and the likelihood of obtaining tertiary education across various age cohorts. The first-stage results, which are statistically significant, suggest that compulsory schooling laws are associated with a substantial positive effect on educational attainment beyond high school. These patterns are observed across all age groups and supported by robust F-statistics, consistent with the relevance of CSL as an instrument for higher education.

The second instrument utilizes the exogenous variation introduced by parental job loss during a child's high school years to investigate how financial disruptions impact long-term wealth accumulation. This approach examines how unexpected financial shocks to a family during critical educational periods influence a child's educational attainment, especially tertiary education, and subsequent wealth. The results of this second instrument are presented in Table 4. The first-stage results show a statistically significant negative effect of parental job loss on college attainment: children whose parents experienced job loss during their high school years are less likely to complete tertiary education, consistent with a liquidity constraint mechanism. The magnitude and significance of this first-stage relationship provide evidence of instrument relevance, though the exclusion restriction, as discussed above, relies on the assumption that job loss does not affect children's long-run wealth through channels other than education. This assumption is strong and cannot be verified directly.

Table 4: I.V. Regression: Parental Job Loss

	Avg	**Cohort**		
		30	**40**	**50 - 60**
Tertiary	19150.10	23802.79*	57676.22*	62512.97
	(19246.83)	(11040.08)	(22440.22)	(53107.25)
First Stage				
PJL → College	−0.069**	−0.085***	−0.062**	0.045
	(0.021)	(0.024)	(0.022)	(0.035)
F-statistic	39.49	26.60	13.30	9.55
Observations	11310	4050	4131	1398

Note: Source: Panel Study of Income Dynamics. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The instrument is parental job loss during the final years of high school. Year and cohort effects are included. Parental wealth is included but not reported for brevity.

The results using parental job loss during high school years as an instrument are broadly consistent with the findings from the first instrument, compulsory schooling laws. The results are suggestive of a positive relationship between education and wealth accumulation, though the caveats regarding the exclusion restriction discussed above apply. Even though the average education effect is not statistically significant, the life cycle effects across different age cohorts are statistically significant, indicating that the impact of education on wealth becomes more pronounced over time.

In terms of tertiary education, the results again show that significant associations are concentrated at higher levels of education. The significance levels are generally lower with parental job loss as the instrument compared to the first instrument. This could be attributed to the more targeted nature of parental job loss, which captures financial disruptions specific to high school years rather than a broader educational policy impact. Despite this, the results remain significant, and are broadly consistent with the IV estimates from compulsory schooling laws. The lower significance levels do not undermine the overall picture; rather, they reflect the more targeted and potentially noisier variation captured by this instrument. Taken together, these results are consistent with the pattern that associations between education and wealth are concentrated at the tertiary level.

As a suggestive falsification exercise, Appendix Table B2 reports a reduced-form regression of wealth on parental job loss using only the non-tertiary sample. Because restricting the sample to non-college households substantially reduces the number of observations and makes cohort-specific estimates unstable, I report only the pooled specification. In that regression, parental job loss is not statistically associated with later wealth once parental wealth, demographic controls, inheritance, cohort effects, and survey-year effects are included. This pattern is consistent with, though it does not establish, the interpretation that the main parental-job-loss results operate primarily through tertiary educational attainment rather than a direct wealth channel.

2.4 Mechanisms

To provide a richer picture of how education may relate to wealth accumulation, this subsection summarizes descriptive evidence on three potential channels: increased productivity, financial literacy, and financial behavior. These patterns are not causal; as with the main results, the regressions are descriptive, and correlates may reflect omitted factors. Their purpose is to characterize patterns in the data that are consistent with the model channels used in the quantitative analysis.

Detailed evidence on each channel is reported in Appendix Tables A6, A7, and A8. The first channel (Table A6) is productivity: education raises labor income, bonuses, and rental returns, all of which attenuate the raw education-wealth gradient, consistent with an income effect (Card, 1999; Gintis, 1971). The second channel (Table A7) is financial literacy: more-educated households are more likely to hold stocks, annuities, and individual retirement accounts (IRAs), and other financial assets (Campbell, 2006; Bertaut, 1998; Bingley & Martinello, 2017), consistent with McKay (2013)'s argument that education improves the ability to search, assess risk, and make effective financial decisions. The third channel (Table A8) is financial behavior: college-educated individuals save more Dynan et al. (2004) and report fewer money problems, consistent with better financial management over the life cycle.

These three channels mirror the transmission mechanisms built into the quantitative model of Section 3: the income premium, education-specific asset returns, and the bequest/savings motive. Importantly, a companion study using the same PSID data and identification strategies, Loaiza (2025), examines the effect of education on financial asset accumulation (savings, annuities/IRAs, and stocks) directly, finding consistent positive effects across all three asset classes for college and postgraduate attainment. That paper corroborates the descriptive correlates here with a richer asset-level analysis, providing additional empirical grounding for the financial literacy and financial behavior channels incorporated in the model.

3 Quantitative Model

After documenting a consistent positive relation between education and net worth, with the most robust patterns at the college and postgraduate levels, a quantitative partial equilibrium life-cycle model is used to study potential scenarios for educational reforms. The empirical findings inform the calibration and design of the model, while the model provides the structure needed to evaluate counterfactual policies that cannot be studied directly in the data. The standard Income Fluctuation Problem is extended by including exogenous connections between education and wealth to create counterfactual scenarios to test these policies.

3.1 Environment

This economy is populated by unitary individuals who live at most T periods, but they also face a positive probability of death π_t starting from retirement at every period. In the first period, agents exogenously acquire the human capital that will affect their working life and retirement. When agents enter the model at age 20, they start their working stage, where they use human capital, consume, and save. Finally, the agents retire at age

65 when they no longer work and only receive interest from accumulated assets, pensions, and utility from consumption.

Preferences of individuals are identical over consumption c_t. These preferences are time separable, with an idiosyncratic stochastic discount factor β_t and survival probabilities s_t at each time t. Additionally, individuals derive utility from leaving a bequest to the next generation.

$$E_0 \left[\sum_{t=0}^{T} \left(\prod_{i=0}^{t} \beta_i \right) s_t \, u(c_t) + (1 - s_t) \, \theta(b_t) \right] \tag{7}$$

Here, s_t is the probability of surviving to period t and $(1 - s_t)$ is the probability of not surviving to period t, leaving a bequest b_t. The period utility function from consumption $u(c_t)$ is of the constant relative risk aversion class, where $\gamma > 1$ is the coefficient of relative risk aversion.

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma} \tag{8}$$

The utility derived from bequests follows De Nardi (2004)

$$\theta^e(b) = \theta_1^e \left(1 + \frac{b}{\theta_2} \right)^{1-\gamma} \tag{9}$$

where θ_1^e is the education-specific bequest strength and θ_2 determines the extent to which bequests are a luxury good. The parameter θ_1^e is allowed to differ between college and non-college agents to capture the well-documented gradient in estate planning and inter-generational wealth transfers by educational attainment (De Nardi, 2004). Calibration details are discussed in Section 3.4.

The initial conditions refer to human capital and assets and differ from agent to agent. Human capital will be provided every period of their working stage of life (from age 20 to 65) to the productive sector. Agents start their life with a level of human capital $h_c \geq 0$ inherited from their parents. Second, the initial level of assets refers to the monetary resources that agents obtained in their first period. These resources can be seen as a regular use of parental wealth. This is assumed to be received at the beginning of their life cycle. Both initial conditions follow a log-normal distribution. The model abstracts from complicated family dynamics and strategic interactions between parents and children and assumes an exogenous intergenerational transmission of human and monetary capital.

At the beginning of the life cycle ($t = 0$), each agent is assigned an education type $e \in \{s, u\}$, where s denotes college-educated (skilled) and u denotes non-college (unskilled). The assignment probability follows the logistic function

$$\Pr(e = s \mid h_c, m_0) = \frac{\exp(\beta_0 + \beta_1 h_c + \beta_2 m_0)}{1 + \exp(\beta_0 + \beta_1 h_c + \beta_2 m_0)}, \tag{10}$$

with coefficients estimated from PSID data (see Appendix B.2), yielding population shares of approximately 34% college-educated and 66% non-college. Education type is fixed for life and enters the model through four channels: (i) a permanent income premium, (ii) education-specific income risk, (iii) mean asset returns, and (iv) the bequest strength parameter. Each channel is described in turn.

The labor income of individuals, y_t, consists of two idiosyncratic components h_t and ξ_t and it is given by the following equation:

$$y_t = h_t \, \xi_t \tag{11}$$

where h_t is a permanent component and ξ_t is a transitory shock. At the start of the working life ($t = 1$), human capital reflects the education premium:

$$h_{e,1} = h_c \cdot \exp(\Delta_e)\,, \qquad \Delta_s = \log(1.40), \quad \Delta_u = 0, \tag{12}$$

so that college graduates enter with a permanent income level 40% above that of non-college agents. This log-shift propagates through the entire earnings path via equation (14).[6]

$$\xi_{t+1} = \begin{cases} \mu & \text{pr } \pi \\ \phi_{t+1}/(1-\pi) & \text{pr } (1-\pi) \end{cases} \tag{13}$$

During all the working stages, labor income is obtained by equation 11. The transitory shock ξ_t, presented in equation 13, gives a small probability π that income will be μ,

[6] The 40% earnings differential at entry is consistent with estimates from Card (1999) and Heckman et al. (2006).

i.e. temporary unemployment or unemployment insurance. Additionally, ϕ is a mean-one IID random variable satisfying $\mathrm{E}_t[\phi_{t+n}] = 1 \ \forall \ n \geq 1$ and $\phi \in [\underline{\phi}, \bar{\phi}]$.

$$h_t = G \ \psi_t \ h_{t-1}, \tag{14}$$

Equation 14 is the permanent component of income: it consists of its previous value, a growth factor G, and a mean-one IID permanent shock ψ_t satisfying $\mathrm{E}_t[\psi_{t+n}] = 1 \ \forall \ n \geq 1$ and $\psi \in [\underline{\psi}, \bar{\psi}]$. Following Meghir and Pistaferri (2004) and Huggett et al. (2011), the shock variances differ by education type:

$$\log \ \psi^e_{t+n} \sim N\Big(-(\sigma^e_\psi)^2/2, \ (\sigma^e_\psi)^2\Big), \tag{15}$$

$$\log \ \phi^e_{t+n} \sim N\Big(-(\sigma^e_\xi)^2/2, \ (\sigma^e_\xi)^2\Big), \tag{16}$$

where $(\sigma^s_\psi, \sigma^s_\xi)$ for college and $(\sigma^u_\psi, \sigma^u_\xi)$ for non-college. The higher volatility for non-college workers reflects stronger exposure to transitory layoffs and occupation-specific demand shocks documented in the empirical earnings literature (Blundell et al., 2008). This implies that non-college agents face a stronger precautionary saving motive, though the resulting precautionary buffer is smaller due to their lower permanent income level. Labor income shocks are independent across agents.[7] This implies that there is no uncertainty over the aggregate labor endowment even though there is uncertainty at the individual level. During retirement, there is no uncertainty from permanent or transitory shocks. Individuals receive an income or pension that is determined by a fixed retirement replacement rate κ obtained from the income of the period before retirement.

It is common in the literature to take the interest rate as fixed, but in this model, the gross return on assets R_t will be state-dependent.[8] This means that there are idiosyncratic rates of return to capital following:

$$\log R^e_t = \bar{u}^e_r + \eta^r_t \ \bar{w}_r \tag{17}$$

where $\bar{w}_r$ is common across groups, η is an IID standard normal innovation process, and

[7] A more complex earning process is provided in De Nardi, Fella, and Paz-Pardo (2020) with a better fit for consumption inequality, but it shows similar results for wealth inequality as a standard process.

[8] For more intuition and theoretical properties on capital income risk and heterogeneous discount factors, check Ma et al. (2020).

$\bar{u}_r^e$ is the education-specific mean log return.[9] College agents face a higher mean return, $\bar{u}_r^s = \log(1.12)$ versus $\bar{u}_r^u = \log(1.04)$ for non-college agents, capturing education-related differences in financial market participation, portfolio sophistication, and investment efficiency documented in Campbell (2006) and Black et al. (2018). Because the mean return differential is compounded over the entire working and retirement life, it is a quantitatively important driver of the widening college–non-college wealth gap in mid and late adulthood. The value function for each education type e is solved separately via the Endogenous Grid Method (EGM), using the education-specific shock nodes (ψ_t^e, ξ_t^e) and mean return $\bar{u}_r^e$, so that agents optimise against the distributions they actually face.
The introduction of discount factors provides additional heterogeneity for individuals in a similar fashion as capital income but with constant values for $\bar{u}_\beta$ as the stationary mean and $\bar{w}_\beta$ as the standard deviation and an IID standard normal innovation process.

$$\log \beta_t = \bar{u}_\beta + \eta_t^\beta \, \bar{w}_\beta \tag{18}$$

The main assumption in this set-up regarding heterogeneous capital risk and discount factors is based on the idea that when R and β were constants, it was required to have $\beta R < 1$ to ensure stability and existence but now that they are stochastic, it is required to fulfill a more general condition:

$$F_{\beta R} := \lim_{n \to T} \left(E \prod_{t=1}^{n} \beta_t \, R_t \right)^{1/n} < 1 \tag{19}$$

The value $F_{\beta R}$ in equation (19) can be thought of as the long run (geometric) average gross rate of return discounted to present value to ensure existence and stability.

3.2 Household Recursive Problem

In this model, a t-year-old agent of education type $e \in \{s, u\}$ chooses consumption c_t and asset holdings a_{t+1} for the next period. The state variables for an agent are the level of human capital h_t, market resources m_t, discount factor β_t, and education type e. Because e is fixed at birth and enters through the parameters $(\Delta_e, \sigma_\psi^e, \sigma_\xi^e, \bar{u}_r^e, \theta_1^e)$, the dynamic

[9] It is possible to improve the model by introducing mean persistence and time-varying volatility to the return on assets highlighted by Fagereng et al. (2016) and Fagereng, Guiso, Malacrino, and Pistaferri (2020).

programming problem is solved separately for each type. The optimal decision rules are functions for consumption, $c^e(h_t, m_t, \beta_t)$, and next-period asset holdings, $a^e(h_t, m_t, \beta_t)$. The household's assets at the end of the period, a_t, are generated from the cash-on-hand m_t (all market resources) minus their consumption c_t, expressed as $a_t = m_t - c_t$. Given this structure, human capital h_t and market resources m_t start with strictly positive values, $(h_t, m_t) \in (0, \infty)$. For simplicity, it is assumed that agents cannot borrow against their future income, implying that they cannot die in debt, conditioned by $c_T \leq m_T$.

During the full-time working stage, from age 20 to 64 (period $t = 1$ to $t = 44$), agents consume, work, and save assets, using their exogenously obtained human capital in the labor market. In this stage, the state variables are presented as a state vector $\bar{z}_t = (h_t, m_t, \beta_t)$. The value function for this period, subject to the previously detailed constraints, is given by:

$$v(\bar{z}_t) = \max_{c_t} \left\{ u(c_t) \ + \ \beta_t s_t E_t \left[v_{t+1}(\bar{z}_{t+1}) \right] + (1 - s_t)\theta(a_t) \right\} \tag{20}$$

s.t.

$$a_t = m_t - c_t \tag{21a}$$

$$y_{t+1} = (\psi_{t+1} \ G \ h_t) \ \xi_{t+1} \tag{21b}$$

$$m_{t+1} = R_{t+1} \ a_t + \ y_{t+1} \tag{21c}$$

During retirement, from age 65 to 90, agents consume, receive their pension, save assets, and face survival probabilities, introducing the risk of death. Consequently, individuals derive utility from leaving bequests to the next generation. The value function for this stage is given by:

$$v(\bar{z}_t) = \max_{c_t} \left\{ u(c_t) \ + \ \beta_t s_t E_t \left[v_{t+1}(\bar{z}_{t+1}) \right] \ + \ (1 - s_t)\theta(a_t) \right\} \tag{22}$$

s.t.

$$a_t = m_t - c_t \tag{23a}$$

$$m_{t+1} = R_{t+1} \ a_t + \ p_{t+1} \tag{23b}$$

3.3 Calibration

The model simulates $n = 100,000$ households, starting work at age 20 and retiring at 65. Each period is one year, with a maximum age of 90, spanning 70 periods. Household preferences use a relative risk aversion coefficient $\gamma = 1.5$ from Attanasio et al. (1999)

and Gourinchas and Parker (2002). One-period survival probabilities s_t come from Bell et al. (1992). The summary of the parameters is presented in Table 5.

Table 5: Summary of Parameters

Parameter	Description	Value
Preferences		
γ	Risk aversion coefficient	1.5
$\bar{u}_\beta$	Stationary mean discount factor	0.91
$\bar{w}_\beta$	Standard deviation discount factor	0.004
θ_1^s	Bequest strength (college)	28.64[a]
θ_1^u	Bequest strength (non-college)	15.91[a]
θ_2	Bequest as luxury good	11.37
Labor Income		
G	Growth income factor	1.03
σ_ψ^s	SD log permanent shock, college	0.08
σ_ψ^u	SD log permanent shock, non-college	0.13
σ_ξ^s	SD log transitory shock, college	0.09
σ_ξ^u	SD log transitory shock, non-college	0.14
Δ_s	Log income premium, college	log(1.40)
π	Probability of zero income shock	0.07
μ	Unemployment insurance payment	0.15
κ	Retirement replacement rate	0.70
Capital Income		
$\bar{u}_r^s$	Mean log return, college	log(1.12)
$\bar{u}_r^u$	Mean log return, non-college	log(1.04)
$\bar{w}_r$	Return volatility (common)	0.215
Initial Conditions		
μ_h	Mean of initial human capital h_p	0.466
σ_h^2	Variance of initial human capital h_p	0.213
μ_a	Mean of initial assets a_p	1.266

[a]Calibrated via SMM to SCF wealth distribution moments (Section 3.4).

The labor income process, based on Carroll et al. (2015), Carroll et al. (1992), and DeBacker et al. (2013), includes an income growth factor $G = 1.03$, reflecting the U.S. average GDP per capita growth rate (1947–2014). The unemployment insurance replacement rate is $\mu = 0.15$ with a probability of $\pi = 0.07$. Pensions during retirement are a fraction $\kappa = 0.70$ of permanent income at retirement. The education-specific shock standard deviations are set to $(\sigma_\psi^s, \sigma_\xi^s) = (0.08, 0.09)$ for college and $(\sigma_\psi^u, \sigma_\xi^u) = (0.13, 0.14)$ for non-college, consistent with the estimates of Meghir and Pistaferri (2004) and Huggett et al. (2011). The baseline non-college mean return to capital is $\exp(\bar{u}_r^u) = 1.04$, with volatility $\bar{w}_r = 0.215$ sourced from Ma et al. (2020). The average discount factor $\bar{\beta} \approx 0.96$ is set by parameters $\bar{u}_\beta = 0.91$ and $\bar{w}_\beta = 0.004$. Details on the discretization process are in Section B.1 in the Appendix.

Probabilities of receiving an inheritance in five-year intervals were derived from PSID data, generating random inheritances to match empirical averages. A parameter was included to reflect that 97% of the population does not receive any inheritance. The model's fit to real data is discussed in Section B.3 in the Appendix. The initial asset distribution uses a Weibull distribution with mean $\mu_m = 0.27955$ from PSID data, including a zero fraction parameter of 0.33 to reflect those with no initial assets. Initial human capital distribution is lognormal, with parameters $\mu_p = 0.23425$ and $\sigma_p = 0.21865$ from PSID data. The common bequest parameter $\theta_2 = 11.37$ is calibrated so that the model replicates the bequest-to-wealth ratio of 1.18, accounting for inter-vivo transfers and college expenditures (De Nardi & Yang, 2016).

3.4 SMM Calibration of Education Channels

The calibration of education-related parameters follows a two-step strategy that separates the distributional calibration from the dynamic channel identification.

Step 1: Distribution calibration. The education-specific bequest strengths (θ_1^s, θ_1^u) are calibrated via SMM to match two moments from the 2019 Survey of Consumer Finances: the overall wealth Gini coefficient (target: 0.82) and the top-10% wealth share (target: 76.7%). These moments are informative about the intensity of bequest motives because stronger bequest preferences concentrate wealth at the top of the distribution, particularly through late-life transfers by college-educated households. The objective is

$$(\hat{\theta}_1^s,\, \hat{\theta}_1^u) = \underset{\theta_1^s, \theta_1^u}{\arg\min} \left[\Big(\text{Gini}(\boldsymbol{\theta}_1) - 0.82\Big)^2 + \Big(S_{10}(\boldsymbol{\theta}_1) - 76.7\Big)^2 \right], \tag{24}$$

where $\text{Gini}(\boldsymbol{\theta}_1)$ and $S_{10}(\boldsymbol{\theta}_1)$ are the model-simulated Gini and top-10% share for a given (θ_1^s, θ_1^u). A grid search over a 6×6 parameter grid followed by Nelder–Mead refinement, using $n = 20{,}000$ simulated agents, yields $\hat{\theta}_1^s = 28.64$ and $\hat{\theta}_1^u = 15.91$.

Step 2: Literature anchors for income and returns. The college income premium $\Delta_s = \log(1.40)$ and the college mean return $\bar{u}_r^s = \log(1.12)$, implying a net return multiplier of approximately $\lambda_R \approx 1.077$ relative to non-college, are fixed at values grounded in the microeconomic literature. The 40% permanent earnings differential is consistent with estimates in Card (1999) and Heckman et al. (2006). The 7.7% annual return differential between college and non-college investors is drawn from Campbell (2006) and Calvet et al. (2007), who document systematic differences in financial market participation, portfolio diversification, and trading efficiency by educational attainment.

This two-step design separates two sources of identification that would otherwise create structural tension in a joint SMM. Bequest parameters, which govern the upper tail of the wealth distribution, are identified from cross-sectional SCF moments that are directly informative about intergenerational transfers. Income and returns parameters, which govern the dynamic age profile of the education-wealth gap, are pinned to direct microeconomic evidence from the earnings and portfolio-returns literatures. The result is a cleaner identification narrative: distributional moments identify bequest motives; the literature identifies the income and returns channels.

Table 6 reports the baseline model's fit to the U.S. wealth distribution. The model replicates the high concentration observed in the data, with a top-10% share of 77.9% against the SCF target of 76.7% and a Gini of 0.84 against the target of 0.82. As an additional validation, Table B3 in the Appendix compares simulated wealth inequality with PSID moments by age group and education level; the model closely matches the observed patterns, especially in mid and late adulthood.

Table 6: Calibration Results: Wealth Distribution

	Avg. Gini	**Percentage Wealth in the Top**						**Bottom**
		1%	5%	10%	20%	40%	60%	40%
U.S. Data 2019	0.82	37.4	65.4	76.7	87.5	96.4	99.7	0.2
Baseline model	0.84	35.1	64.4	77.9	89.0	96.2	98.5	1.5

Source for U.S. Data: Survey of Consumer Finances, 2019. Model: main simulation (income + returns), n = 100,000 agents, bequest parameters $\theta_1^s = 28.64$, $\theta_1^u = 15.91$ calibrated to SCF Gini and top-10% share.

3.5 Decomposition Simulations and Policy Counterfactuals

The simulation design proceeds in three steps. The first isolates the channels through which education generates the age-gradient in the IV estimates. The second quantifies the income-risk mechanism and connects it to the financial behaviour channel documented in Loaiza (2025). The third evaluates whether returns-based or access-oriented policies can meaningfully close the gradient. All simulations share the same baseline college-assignment probabilities; policy interventions act post-education. The key diagnostic is the age-gradient ratio $\hat{\beta}_{50s}/\hat{\beta}_{30s}$, computed from the same IHS regression used in Section 2, which captures how fast the education-wealth gap accelerates over the life cycle. Table 7 summarises all results.

3.5.1 Channel Decomposition

Income Channel. College graduates receive the income premium $\Delta_s = \log(1.40)$ but face the same mean asset return $\bar{u}_r^u$ as non-college agents. This isolates the pure income-compounding channel: a permanently higher earnings path translates into higher savings each period, widening the wealth gap through repeated deposits rather than any difference in investment performance. The result is striking: income alone produces a *negative* age-gradient ($\hat{\beta}_{50s}/\hat{\beta}_{30s} = -0.13\times$, accounting for -137% of the data gradient). The mechanism is precautionary saving: non-college agents, facing higher income risk and lower permanent income, save relatively more aggressively in their 30s, compressing the early-career wealth gap. This early compression means that income by itself cannot account for the observed age-acceleration.

Returns Channel. College graduates receive no income premium but face the higher mean return $\bar{u}_r^s = \log(1.12)$. This isolates the compounding-returns channel: even with identical earnings, a higher mean return amplifies assets multiplicatively each period, producing a gap that is convex in age and accelerates disproportionately in late adulthood. The returns channel alone generates a gradient of $1.48\times$, accounting for 59% of the empirical age-acceleration, nearly the full observed pattern.

Main Model. Both premiums are active simultaneously. The full model produces a gradient of $1.45\times$ (54% of the data), slightly below the returns-only simulation because the income-precautionary mechanism partially offsets the compounding effect early in the life cycle. The Main Model is the calibrated benchmark for all subsequent comparisons. The key finding of the decomposition is that differential returns to capital, not income differences, are the primary driver of the age-acceleration in the education-wealth gap.

3.5.2 The Income Risk Mechanism

The income shock volatility of non-college agents is reduced to college levels: $(\sigma_\psi^u, \sigma_\xi^u)$ are set from $(0.13, 0.14)$ to $(0.08, 0.09)$. The EGM is re-solved with the lower volatility so that non-college agents optimally adjust their precautionary saving behaviour. College agents retain both the income premium and the returns premium. This simulation asks: if non-college workers had the same income stability as college workers, how much of the age gradient would remain?

The result quantifies the financial behaviour channel identified in Loaiza (2025), who documents that education reduces financial problems, improves money management, and lowers exposure to income disruptions. In the life-cycle model, these effects translate directly into lower effective income volatility for college graduates. The simulation finds a gradient of 1.27× (32% of the data), compared to 1.45× (54%) in the Main Model. Equalising income risk, therefore, eliminates approximately 22 percentage points of the age gradient, isolating the quantitative contribution of the financial behaviour channel. The three mechanisms identified in Loaiza (2025), income level, financial behaviour, and risk tolerance, map directly onto the Income Channel, Income Risk Equalization, and Returns Channel simulations respectively, providing a model-based decomposition of the companion empirical findings. The general equilibrium implications of this financial literacy channel are further studied by Kim (2024), who shows that closing the financial knowledge gap substantially compresses wealth inequality in a life-cycle portfolio choice model.

3.5.3 Policy Counterfactuals

Financial Democratization. Non-college agents gain full access to the college mean return $\bar{u}_r^s$, so that all agents face the same return distribution. The non-college EGM is resolved with the higher mean return. College agents still receive the income premium, so the remaining gap is driven entirely by earnings. This simulation provides an upper bound on returns-equalisation policy: it produces a gradient of $-2.77\times$, indicating that fully equalising returns reverses the age-ordering of the gap. The policy over-corrects because non-college agents, who start with lower wealth, benefit disproportionately from compound returns once the disadvantage is removed. This result confirms the primacy of the returns channel: it is large enough that equalising it entirely flips the direction of the age gradient.

Financial Education. Non-college agents receive a partial returns improvement equal to 10% of the college–non-college return differential, so their effective mean return becomes $1+0.10\times(\lambda_R-1) \approx 1.008$. These models targeted financial education or investment literacy programmes that close a fraction of the returns gap without full democratization. The result is a gradient of 1.51× (62% of data), slightly above the Main Model. The mild increase reflects an early-compounding paradox: a small improvement in non-college returns in early adulthood raises the young-cohort wealth gap (the $\hat{\beta}_{30s}$ denominator) by proportionally less than it raises the mid-adulthood gap, steepening the ratio. Contrasting this with the full Financial Democratization, which over-corrects, and the Main Model, which under-explains, suggests that a programme closing roughly 40–50% of the returns gap would match the data gradient.

Education Expansion. The college share is raised from 34% to 37.4%, a 10% increase in the number of college graduates. All other parameters, including the income and returns premiums, are held at their calibrated values. This simulation asks whether broadening access to college can compress the aggregate education-wealth gradient. The result is a gradient of 1.45× (54%), virtually identical to the Main Model. Education expansion does not change the per-person dynamics of the gap: marginal entrants face the same income premium, returns premium, and life-cycle compounding as existing college graduates. The aggregate gradient is unchanged because the per-person mechanisms are unchanged. This finding reinforces the conclusion that policies targeting the returns channel, rather than the quantity of college graduates, are more likely to compress the age-acceleration of wealth inequality.

4 Discussion

The main implication of the quantitative results is that the education-wealth gap is not primarily an earnings phenomenon. The empirical estimates show that the college premium in wealth widens sharply with age, and the model indicates that this widening is driven mainly by financial compounding. In the decomposition, the returns channel alone accounts for nearly 60% of the empirical age gradient, while the income channel by itself fails to reproduce the observed acceleration. This pattern suggests that the long-run wealth advantage associated with tertiary education arises less from higher annual deposits and more from the rate at which assets compound once they are accumulated. In that sense, the relevant policy problem is not only who earns more, but who has access to the returns process that governs wealth growth later in life.

The income-risk mechanism adds an important layer to this interpretation. Equalizing non-college income volatility to college levels lowers the age gradient from 1.45× in the

Table 7: education-wealth Age-Gradient: Channel Decomposition and Policy Counterfactuals

Simulation	College coeff. $\hat{\beta}_{as}$ (IHS units) 30s	40s	50s	60s	Grad.	% Data
(a) Channel decomposition						
Income Channel	−0.320	−0.201	0.042	0.306	−0.13×	−137%
Returns Channel	1.222	1.542	1.814	2.184	1.48×	59%
Main Model	1.258	1.558	1.823	2.193	1.45×	54%
(b) Income risk mechanism						
Income Risk Equalization	1.651	1.961	2.090	2.298	1.27×	32%
(c) Policy counterfactuals						
Financial Democratization	−0.038	0.024	0.105	0.228	−2.77×	−455%
Financial Education (+10% of gap)	1.060	1.334	1.602	1.970	1.51×	62%
Education Expansion (+10% share)	1.262	1.562	1.825	2.191	1.45×	54%
CSL IV (data)		1.83×	100%			

Note: $\hat{\beta}_{as}$ is the college coefficient from IHS(net worth) $= \beta \cdot \text{College} + \gamma \cdot p_0 +$ period FE, estimated on model-simulated panel data. Gradient $= \hat{\beta}_{50s}/\hat{\beta}_{30s}$ (scale-free). % Data = (Gradient − 1)/(Data gradient − 1) × 100. Data gradient from CSL IV (Table 3): 38,114/20,843 = 1.83×. Model $\hat{\beta}$ are in IHS model units; data $\hat{\beta}$ are in USD, the gradient ratio is directly comparable across both. Income Risk Equalization sets $(\sigma^u_\psi, \sigma^u_\xi)$ to college levels $(0.08, 0.09)$ and re-solves the EGM. Education Expansion raises the college share from 34% to 37.4%. Financial Education sets the non-college return multiplier to $1 + 0.10 \times (\lambda_R - 1) \approx 1.008$.

Main Model to 1.27×, implying that part of the education-wealth gap reflects differences in financial stability and precautionary saving conditions rather than differences in earnings or asset returns alone. In the model, lower volatility allows households to save under less pressure from short-run shocks, changing the timing of accumulation over the life cycle. The broader lesson is that education shapes wealth through a bundle of mechanisms, income, returns, and financial stability, that reinforce one another over time.

The policy counterfactuals clarify which margins appear most consequential. Full Financial Democratization more than closes the observed gradient, indicating that equalizing returns is a quantitatively powerful but extreme benchmark. By contrast, a modest Financial Education intervention that closes only 10% of the return gap leaves the gradient above the Main Model, while a 10% increase in the college share leaves it essentially unchanged. These results suggest that marginal enrollment expansion, by itself, does little to compress the per-person dynamics of wealth accumulation once the financial environment is held fixed. Policies that alter post-schooling financial conditions, such as access to high-return assets (Lusardi & Mitchell, 2023), participation in tax-advantaged savings vehicles, or the ability to avoid costly portfolio mistakes (Kim, 2024), appear more powerful for reducing the age-acceleration of the education-wealth gap than policies that simply increase the number of college graduates at the margin.

Several limitations should be kept in mind. First, parental wealth and education proxy for intergenerational advantage but do not capture all informal transfers, social networks, or cultural capital that may affect both schooling and wealth accumulation. Second, the IV evidence remains suggestive: compulsory schooling laws and parental job loss provide useful variation, but their exclusion restrictions cannot be verified directly. Third, the model is partial equilibrium and therefore abstracts from wage, price, or asset-market adjustments that could accompany large-scale policy reforms. Finally, some policy counterfactuals, especially full Financial Democratization, should be read as upper-bound experiments rather than literal predictions. These limitations qualify the quantitative magnitudes, but they do not overturn the paper's central message that financial channels are crucial to understanding how education translates into wealth over the life cycle.

Future research could extend this framework in several directions. One is to incorporate earlier educational investments and study how pre-college skill formation affects later wealth dynamics. Another is to allow for richer financial heterogeneity, including participation constraints, heterogeneous fees, or persistence in returns, which could sharpen the policy interpretation of the returns channel. A third is to bring administrative or linked intergenerational data to bear on transfers, inheritances, and the timing of educational decisions. These extensions would help distinguish more precisely between educational attainment, financial capability, and family resources as sources of long-run wealth inequality.

5 Conclusions

This paper examined whether education shapes wealth accumulation over the life cycle and through which mechanisms the education-wealth gap widens with age. Using three complementary empirical strategies with PSID data, the analysis provides consistent evidence of a positive relation between education and wealth, concentrated at the tertiary level. Crucially, the gap is not constant: it grows markedly from the 30s to the 60s, consistent with cumulative compounding rather than a one-time earnings shift.

A quantitative life-cycle model decomposes this age-acceleration into distinct channels.

The central result is that differential returns to capital, not income differences alone, drive the widening gap. The returns channel alone accounts for nearly 60% of the empirical age gradient, while the income channel by itself produces the wrong sign. An income-risk mechanism adds a further layer: equalizing non-college income volatility materially compresses the gradient, implying that financial stability and precautionary saving conditions are also part of the education effect on wealth.

The policy implications follow directly. Policies that equalize financial returns across education groups have much larger effects on the age gradient than expanding college enrollment: full financial democratization more than closes the gradient, while a 10% increase in the college share leaves it essentially unchanged. The broader implication is that the distributional consequences of education policy depend on which margin is targeted. Policies aimed at financial access and post-schooling wealth-building conditions may be more effective at compressing the education-wealth gap than enrollment expansion alone.

Data Availability

The data used in this study come from the Panel Study of Income Dynamics (PSID), which is publicly available upon registration from the PSID website. All data sources are described in detail in the paper and its appendix. Replication code will be made available upon publication.

Declaration of Competing Interests

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

Acemoglu, D., & Angrist, J. (2000). How large are human-capital externalities? evidence from compulsory schooling laws. *NBER macroeconomics annual*, *15*, 9–59.

Alvaredo, F., Chancel, L., Piketty, T., Saez, E., & Zucman, G. (2018). *World inequality report 2018*. Belknap Press.

Atkinson, A. B., & Bourguignon, F. (2014). *Handbook of income distribution* (Vol. 2). Elsevier.

Attanasio, O. P., Banks, J., Meghir, C., & Weber, G. (1999). Humps and bumps in lifetime consumption. *Journal of Business & Economic Statistics*, *17*(1), 22–35.

Audoly, R., McGee, R., Ocampo, S., & Paz-Pardo, G. (2024). *The life-cycle dynamics of wealth mobility* (Staff Report No. 1097). Federal Reserve Bank of New York.

Barth, D., Papageorge, N. W., Thom, K., & Velásquez-Giraldo, M. (2023). *Genetic endowments, income dynamics, and wealth accumulation over the lifecycle* (Working Paper No. 30350). National Bureau of Economic Research. doi: 10.3386/w30350

Bell, F. C., Wade, A. H., & Goss, S. C. (1992). *Life tables for the united states social security area, 1900-2080* (No. 107). US Department of Health and Human Services, Social Security Administration ....

Benhabib, J., Bisin, A., & Luo, M. (2019). Wealth distribution and social mobility in the us: A quantitative approach. *American Economic Review*, *109*(5), 1623–47.

Bertaut, C. C. (1998). Stockholding behavior of us households: Evidence from the 1983–1989 survey of consumer finances. *Review of Economics and Statistics*, *80*(2), 263–275.

Bingley, P., & Martinello, A. (2017). The effects of schooling on wealth accumulation approaching retirement. *Department of Economics, Lund University Working Papers*, *2017*(9).

Black, S. E., Devereux, P. J., Lundborg, P., & Majlesi, K. (2015). *Poor little rich kids? the role of nature versus nurture in wealth and other economic outcomes and behaviors* (Tech. Rep.). National Bureau of Economic Research.

Black, S. E., Devereux, P. J., Lundborg, P., & Majlesi, K. (2018, 02). Learning to Take Risks? The Effect of Education on Risk-Taking in Financial Markets*. *Review of Finance*, *22*(3), 951-975. Retrieved from `https://doi.org/10.1093/rof/rfy005` doi: 10.1093/rof/rfy005

Blanden, J., & Machin, S. (2004). Educational inequality and the expansion of uk higher education. *Scottish Journal of Political Economy*, *51*(2), 230–249.

Blundell, R., Pistaferri, L., & Preston, I. (2008). Consumption inequality and partial insurance. *American Economic Review*, *98*(5), 1887–1921.

Calvet, L. E., Campbell, J. Y., & Sodini, P. (2007). Down or out: Assessing the welfare costs of household investment mistakes. *Journal of Political Economy*, *115*(5), 707–747. doi: 10.1086/524204

Campbell, J. Y. (2006). Household finance. *The journal of finance*, *61*(4), 1553–1604.

Card, D. (1999). The causal effect of education on earnings. In *Handbook of labor economics* (Vol. 3, pp. 1801–1863). Elsevier.

Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics letters*, *91*(3), 312–320.

Carroll, C. D., Hall, R. E., & Zeldes, S. P. (1992). The buffer-stock theory of saving:

Some macroeconomic evidence. *Brookings papers on economic activity*, *1992*(2), 61–156.

Carroll, C. D., Slacalek, J., & Tokuoka, K. (2015). Buffer-stock saving in a krusell–smith world. *Economics Letters*, *132*, 97–100.

Charles, K. K., & Hurst, E. (2003). The correlation of wealth across generations. *Journal of political Economy*, *111*(6), 1155–1182.

Charles, K. K., Hurst, E., & Notowidigdo, M. J. (2018). Housing booms and busts, labor market opportunities, and college attendance. *The American Economic Review*, *108*(10), 2947–2994.

Chevalier, A., Harmon, C., O'Sullivan, V., & Walker, I. (2013). The impact of parental income and education on the schooling of their children. *IZA Journal of Labor Economics*, *2*(1), 8.

Daminato, C., & Pistaferri, L. (2024). *Returns heterogeneity and consumption inequality over the life cycle* (Working Paper No. 32490). National Bureau of Economic Research.

DeBacker, J., Heim, B., Panousi, V., Ramnath, S., & Vidangos, I. (2013). Rising inequality: transitory or persistent? new evidence from a panel of us tax returns. *Brookings Papers on Economic Activity*, *2013*(1), 67–142.

De Nardi, M. (2004). Wealth inequality and intergenerational links. *The Review of Economic Studies*, *71*(3), 743–768.

De Nardi, M., Fella, G., & Paz-Pardo, G. (2020). Nonlinear household earnings dynamics, self-insurance, and welfare. *Journal of the European Economic Association*, *18*(2), 890–926.

De Nardi, M., & Yang, F. (2016). Wealth inequality, family background, and estate taxation. *Journal of Monetary Economics*, *77*, 130–145.

Dynan, K. E., Skinner, J., & Zeldes, S. P. (2004). Do the rich save more? *Journal of political economy*, *112*(2), 397–444.

Fagereng, A., Guiso, L., Holm, M. B., & Pistaferri, L. (2020). *K-Returns to Education* (EIEF Working Papers Series No. 2002). Einaudi Institute for Economics and Finance (EIEF). Retrieved from `https://ideas.repec.org/p/eie/wpaper/2002.html`

Fagereng, A., Guiso, L., Malacrino, D., & Pistaferri, L. (2016). Heterogeneity in returns to wealth and the measurement of wealth inequality. *American Economic Review*, *106*(5), 651–55.

Fagereng, A., Guiso, L., Malacrino, D., & Pistaferri, L. (2020). Heterogeneity and persistence in returns to wealth. *Econometrica*, *88*(1), 115–170.

Gintis, H. (1971). Education, technology, and the characteristics of worker productivity. *The American Economic Review*, *61*(2), 266–279.

Girshina, A. (2019). *Wealth, savings, and returns over the life cycle: the role of education* (Tech. Rep.). Working Paper.

Gourinchas, P.-O., & Parker, J. A. (2002). Consumption over the life cycle. *Econometrica*, *70*(1), 47–89.

Heckman, J. J., Lochner, L., & Todd, P. E. (2006). Earnings functions, rates of return and treatment effects: The mincer equation and beyond. *Handbook of the Economics of Education*, *1*, 307–458.

Huggett, M., Ventura, G., & Yaron, A. (2011). Sources of lifetime inequality. *American Economic Review*, *101*(7), 2923–54.

Karagiannaki, E. (2017). The effect of parental wealth on children's outcomes in early adulthood. *The Journal of Economic Inequality*, *15*(3), 217–243.

Keller, K. R. (2010). How can education policy improve income distribution? an empirical analysis of education stages and measures on income inequality. *The Journal of Developing Areas*, 51–77.

Kim, M. (2024). *Financial literacy, portfolio choice, and wealth inequality: A general equilibrium approach* (Working Paper No. 2024-4). Wharton Pension Research Council.

Lavecchia, A. M., Liu, H., & Oreopoulos, P. (2016). Behavioral economics of education: Progress and possibilities. In *Handbook of the economics of education* (Vol. 5, pp. 1–74). Elsevier.

Loaiza, F. (2025). Implications of schooling on financial assets. *Education Economics*, *0*(0), 1–16. Retrieved from https://doi.org/10.1080/09645292.2025.2607110 doi: 10.1080/09645292.2025.2607110

Lochner, L. (2010). Education policy and crime. In *Controlling crime: strategies and tradeoffs* (pp. 465–515). University of Chicago Press.

Lusardi, A., & Mitchell, O. S. (2023). The importance of financial literacy: Opening a new field. *Journal of Economic Perspectives*, *37*(4), 137–154.

Ma, Q., Stachurski, J., & Toda, A. A. (2020). The income fluctuation problem and the evolution of wealth. *Journal of Economic Theory*, *187*, 105003.

McKay, A. (2013). Search for financial returns and social security privatization. *Review of Economic Dynamics*, *16*(2), 253–270.

Meghir, C., & Pistaferri, L. (2004). Income variance dynamics and heterogeneity. *Econometrica*, *72*(1), 1–32.

Mincer, J. (1958). Investment in human capital and personal income distribution. *Journal of political economy*, *66*(4), 281–302.

Psacharopoulos, G., & Patrinos, H. A. (2018). *Returns to investment in education: a decennial review of the global literature*. The World Bank.

Saez, E., & Zucman, G. (2016, 02). Wealth Inequality in the United States since 1913: Evidence from Capitalized Income Tax Data *. *The Quarterly Journal of Economics*, *131*(2), 519-578. Retrieved from https://doi.org/10.1093/qje/qjw004 doi: 10.1093/qje/qjw004

Stevens, A. H., & Schaller, J. (2011). Short-run effects of parental job loss on children's academic achievement. *Economics of Education Review*, *30*(2), 289–299.

A Econometric Analysis: Additional Information

A.1 Description and Summary of Variables

Table A1: Description of Variables

Variable	Description
Wealth	Total value of financial assets, non-financial assets, less the value of liabilities (mortgage and land contracts, family mortgage debt, education debt owed for personal and government loans, and other debt), and excluding the value of home equity.
Education	Highest year of education completed. Education is classified into 5 categories (detailed in subsection A.3).
Par. Wealth	Parental net worth reported when the child was young.
Par.Education W.	Highest year of education completed by the mother.
Par.Education H.	Highest year of education completed by the father.
Par. Income	Total parental income reported when the child was young.
Ability	IQ score tests as a proxy for ability with results that range from zero to thirteen.
Parents	Reports as “1" if the individual lived with both parents until 16 years old and “0" otherwise.
Inheritance	Value of inheritance received by the individual.
Age	Current age of each individual in a particular year.
Race	*Race* is reported as "1" if White and "0" for others.
Sex	*Sex* is reported as "1" for males and "0" for females.
Compulsory Schooling Laws (CSL)	Compulsory schooling laws are the minimum years of education that an individual had as law in a respective state when 14 years of age.
Parental Job Loss (PJL)	Parental job loss is calculated by summing the hours of unemployment for each parent during the years when the child is between 15 and 18 years of age.

Table A2: Classification of the Educational Variable

Level	Year	Pct.
High school D.O.	0-11	15.1
High school	12	32.7
College	13-14	20.2
College	15-16	20.4
Post-graduate	17	11.6

Source: Panel Study of Income Dynamics Data

Table A3: Summary Statistics

	Summary Statistics				
	Obs.	Mean	St.D.	Min	Max
Age	7486	50.97	8.44	30	70
Sex	7486	0.76	0.42	0	1
Race	7486	0.83	0.38	0	1
Parents	7486	0.81	0.39	0	1
Ability	7486	9.76	2.08	0	13

Note: Source: Panel Study of Income Dynamics. Significance levels are denoted as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Data in this analysis is used with sampling weights.

A.2 Inverse Hyperbolic Sine Transformation of Wealth Variable

Household net worth in the PSID exhibits substantial right-skewness and includes zero and negative values. To accommodate these features while retaining all observations, the analysis applies an inverse hyperbolic sine (IHS) transformation to net worth. The IHS transformation preserves non-positive values and provides a log-like transformation for large values without requiring strict positivity. Formally, the inverse hyperbolic sine transformation is defined as:

$$\operatorname{asinh}(x) = \ln! \left(x + \sqrt{x^2 + 1} \right). \tag{A1}$$

A generalized, scaled version of the transformation is given by:

$$\mathrm{IHS}_\lambda(x) = \frac{1}{\lambda} \operatorname{asinh}(\lambda x) = \frac{1}{\lambda} \ln! \left(\lambda x + \sqrt{(\lambda x)^2 + 1} \right), \tag{A2}$$

where $\lambda > 0$ is a scaling parameter. In the baseline analysis, $\lambda = 0.0001$ is used to improve numerical stability given the wide range of net worth values. As a robustness check, specifications using $\lambda = 1$ yield qualitatively similar results.

A.3 Descriptive Analysis

Table A4: Mean Wealth by Education and Cohort

	Education Level				
Age Cohort	**0**	**1**	**2**	**3**	**4**
30	4152.8	35925.2	61618.8	268526.7	97615.5
	(600.0)	(7500.0)	(12000.0)	(55625.0)	(38200.0)
40	15888.6	55392.5	79684.1	658240.8	239787.6
	(200.0)	(10700.0)	(17000.0)	(105000.0)	(103000.0)
50	38455.3	103747.9	115654.5	817157.1	463874.9
	(1600.0)	(13014.0)	(26750.0)	(152000.0)	(218500.0)
60	39604.4	159808.1	220448.7	831762.3	909346.1
	(3300.0)	(13000.0)	(56000.0)	(264000.0)	(360300.0)

Note: Source: Panel Study of Income Dynamics, 1999 to 2019. The median value in parentheses. The data is used with sampling weights.

A.4 Additional Results: Instrumental Variables

Table A5: I.V. Regression: Compulsory Schooling Laws

(a) Avg. Education					
	Avg	**Cohort**			
		30	**40**	**50**	**60**
Education	6155.57**	3977.16*	6171.54***	7609.57***	11040.17***
	(2189.31)	(1964.08)	(1246.49)	(1437.43)	(2826.96)
F-statistic	38.58	17.02	51.70	53.82	21.98
Observations	10281.00	1389.00	3912.00	3681.00	1243.00
(b) College Education					
	Avg	**Cohort**			
		30	**40**	**50**	**60**
College	48225.00+	44545.21	39592.40***	61499.81***	71930.67**
	(26147.80)	(30959.81)	(9109.58)	(15707.51)	(27347.92)
F-statistic	22.21	8.59	39.84	29.44	9.97
Observations	10281.00	1389.00	3912.00	3681.00	1243.00
(c) Postgraduate Education					
	Avg	**Cohort**			
		30	**40**	**50**	**60**
Postgraduate	75971.12	39170.62+	97973.17**	100234.87**	1192573.86
	(56081.63)	(22383.26)	(37598.58)	(34781.66)	(6269276.76)
F-statistic	15.45	12.71	14.32	15.95	0.05
Observations	10281.00	1389.00	3912.00	3681.00	1243.00

Note: Source: Panel Study of Income Dynamics. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The instrument is the years of compulsory schooling by state. Year and cohorts effects are included. Parental wealth is included but not reported for brevity.

A.5 Additional Results: Mechanism Channels

The three tables below provide the full regression evidence on the mechanism channels discussed in Section 2.4. Table A6 documents the productivity channel (labor income, bonuses, and rent). Table A7 documents the financial literacy channel (stocks, annuities/IRAs, other financial assets, and interest income). Table A8 documents the financial behavior channel (savings and money problems). All regressions use the same specification and sample as the main empirical analysis.

Table A6: Wealth's Regression Mechanisms: Productivity Effect

Dependent Variable: Wealth			
	(A)	**(B)**	**(C)**
High school	1027.58^{+}	1207.42*	1253.42*
	(600.53)	(609.71)	(607.00)
Some College	2126.64**	2414.61***	2469.98***
	(673.69)	(675.49)	(672.13)
College	1771.38*	2372.60**	2464.87**
	(782.51)	(781.74)	(777.57)
Postgraduate	1523.49	2457.52*	2581.73**
	(989.47)	(983.72)	(981.38)
Labor Income	0.15***		
	(0.02)		
Bonuses		0.23***	
		(0.04)	
Rent			0.30***
			(0.04)
Adjusted R^2	0.28	0.26	0.27
Observations	20558	20558	20558

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are heteroskedastic robust. Time, socio-demographics, inheritance, parental education and wealth, and cohort effects are included. Socio-demographic variables include age, sex, and race of individuals. The constant term is included but not reported for brevity.

Table A7: Wealth's Regression Mechanisms: Financial Literacy

	Dependent Variable: Wealth			
	(A)	**(B)**	**(C)**	**(D)**
High school	1371.04*	1591.78**	1050.28+	1224.15*
	(571.21)	(531.46)	(567.45)	(608.22)
Some College	2223.07***	2346.42***	2184.52***	2430.21***
	(624.32)	(591.13)	(643.66)	(674.83)
College	1726.54*	1563.35*	2251.57**	2452.27**
	(730.65)	(684.46)	(749.00)	(780.23)
Postgraduate	1360.07	364.85	2051.26*	2583.78**
	(913.79)	(866.00)	(945.17)	(984.42)
Stocks	0.48***			
	(0.01)			
Annuity/IRA		0.57***		
		(0.01)		
Other Assets			0.51***	
			(0.02)	
Interest				0.09***
				(0.02)
Adjusted R^2	0.38	0.45	0.32	0.27
Observations	20558	20558	20558	20558

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are heteroskedastic robust. Time, socio-demographics, inheritance, parental education and wealth, and cohort effects are included. Socio-demographic variables include age, sex, and race of individuals. The constant term is included but not reported for brevity.

Table A8: Wealth's Regression Mechanisms: Financial Behavior

Dependent Variable: Wealth		
	(A)	**(B)**
High school	1277.80*	949.33
	(508.51)	(613.50)
Some College	1828.73**	2062.60**
	(571.34)	(690.39)
College	1323.68*	2044.10*
	(658.18)	(794.41)
Postgraduate	382.32	2084.18*
	(874.28)	(995.52)
Savings	0.78***	
	(0.02)	
Money Problem		−4768.88***
		(576.23)
Adjusted R^2	0.46	0.28
Observations	18057	19929

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are heteroskedastic robust. Time, socio-demographics, inheritance, parental education and wealth, and cohort effects are included. Socio-demographic variables include age, sex, and race of individuals. The constant term is included but not reported for brevity.

A.6 Additional Results: Parental Wealth versus Income

The regression results in Table A9 in the Appendix compare the effects of different parental economic background variables on education estimates and other control variables, focusing on parental income and wealth. Parental income has a significant effect on a child's future outcomes, but it is not as strong as parental wealth. Following previous results, this analysis focuses on college and postgraduate-educated individuals when the head of the family unit was young. Column (A) of Table A9 includes parental income, while column (B) includes parental wealth. The estimates that account for parental wealth are more attenuated than those that use parental income.

Key comparisons show that an additional unit of parental income increases the child's future wealth by 21%, whereas an increase in parental wealth generates a 28% increase in future wealth. These findings suggest that parental wealth has a greater impact on a child's life outcomes than parental income. Including parental income or wealth helps better estimate the effect of education. The coefficient for education is lower when parental wealth is considered, indicating that only considering parental income might overestimate the effect of education on wealth.

Table A9: Parental Income and Wealth

Dependent Variable: Wealth		
	(A)	**(B)**
High school	1211.87*	1220.52*
	(616.73)	(611.31)
Some College	2350.31***	2429.58***
	(678.62)	(677.35)
College	2492.54**	2439.55**
	(781.36)	(783.29)
Postgraduate	2751.91**	2606.89**
	(1005.75)	(988.58)
Inheritance	0.16***	0.15***
	(0.02)	(0.02)
Par.Education W.	571.10*	362.50
	(251.78)	(254.99)
Par.Education H.	910.48***	597.69*
	(269.70)	(266.59)
Parental Income	0.21***	
	(0.04)	
Parental Wealth		0.28***
		(0.02)
Adjusted R^2	0.24	0.26
Observations	20461	20558

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are heteroskedastic robust. Time, socio-demographics, and cohort effects are included. Socio-demographic variables include age, sex, and race of individuals. The constant term is included but not reported for brevity.

A.7 Quantile Regression

This analysis is introduced after the main relationship has been explored. It is done with the same data and covariates, and under a similar specification as the first empirical strategy. The quantile regression follows

$$Q_q(W_{it}) = \alpha_q + \beta_{0q}\, Educ_i + \beta_{1q}\, X_i + \beta_{2q}\, SD_{it} + \gamma_t + \upsilon_{itq} \tag{A3}$$

where the equation A3 is jointly estimated for the 10th, 25th, 50th, 75th, 95th, and 99th percentiles of the distribution of the wealth. The quantile regression, in contrast to the RE regression of equation 1, aims to explore the non-linear effects of education on wealth accumulation to see if education affects specific parts of the distribution differently. This regression also provides results by age cohorts to observe effects at different stages of life and by educational categories.

The results obtained in table A10 show positive and statistically significant coefficients for the education categories not only for the average but also over the life cycle. The clear results show that for college graduates, there is no effect, and for postgraduate educated individuals, there is a negative effect of education on wealth when these individuals belong to the 10th percentile of the wealth distribution. The effects of education for the higher percentiles increase until a peak point between the 50th and 75th percentile when later, the coefficients start reducing their value. Similar non-linear effects can also be seen for variables such as inheritance and parental wealth. These results might suggest that even though these variables contribute to wealth accumulation for the majority of individuals, there are other more important influential factors than those on top of the wealth distribution. These estimates obtained from the quantile regression can be appreciated more clearly in the figure A1, which additionally reports the OLS results with a dashed line and confidence intervals with a dotted line. The non-linear effects are seen for education, inheritance, and parental wealth.

Table A10: Quantile Regression: Effects of Education on Wealth

	(A) Quantiles of Wealth Distribution					
	0.10	**0.25**	**0.50**	**0.75**	**0.95**	**0.99**
Highschool	2233.98***	906.30*	3472.00***	4739.12***	3829.28**	7582.75***
	(557.24)	(430.68)	(486.34)	(559.06)	(1212.89)	(879.15)
Some College	431.57	1336.53*	5652.41***	7088.34***	6250.97***	8841.04***
	(593.41)	(576.08)	(546.04)	(623.92)	(1174.98)	(1084.64)
College	850.80	5178.97***	10924.69***	12501.78***	10550.40***	14881.19***
	(924.38)	(731.30)	(590.12)	(606.71)	(1204.03)	(1629.12)
Postgraduate	−6113.21***	5677.32***	14522.20***	14932.71***	11275.85***	12084.83***
	(1232.35)	(1203.41)	(728.57)	(650.08)	(1170.62)	(1143.66)
Inheritance	0.24**	0.41***	0.35***	0.21***	0.08***	0.05
	(0.08)	(0.05)	(0.02)	(0.02)	(0.02)	(0.04)
Parental Wealth	0.17***	0.23***	0.28***	0.27***	0.20***	0.03
	(0.02)	(0.02)	(0.01)	(0.01)	(0.02)	(0.03)
Observations	20556	20556	20556	20556	20556	20556

	(B) Quantiles of Wealth Distribution by Age Cohort					
	Cohort: 40			**Cohort: 60**		
	0.25	**0.50**	**0.95**	**0.25**	**0.50**	**0.95**
High school	3288.23***	2976.58***	4647.41**	4422.23***	2063.75	−1332.98^{+}
	(428.65)	(551.23)	(1699.36)	(1080.78)	(1311.62)	(764.11)
Some College	3939.58***	4930.85***	8209.59***	7379.38***	10804.34***	2450.88*
	(628.42)	(651.10)	(1767.93)	(1161.26)	(1977.45)	(1109.39)
College	7845.91***	11137.99***	12725.15***	12094.71***	12963.23***	7232.86***
	(704.80)	(774.17)	(1810.63)	(1452.85)	(1613.91)	(1006.09)
Postgraduate	4110.27*	10966.64***	12692.98***	22372.68***	20440.86***	7044.63***
	(1824.02)	(938.93)	(1852.72)	(1108.08)	(2081.58)	(903.51)
Inheritance	0.68***	0.67***	0.15***	0.83***	0.42***	0.24***
	(0.10)	(0.05)	(0.04)	(0.06)	(0.08)	(0.07)
Parental Wealth	0.21***	0.27***	0.13***	0.09**	0.15**	0.19***
	(0.02)	(0.02)	(0.03)	(0.03)	(0.05)	(0.03)
Observations	6436	6436	6436	1920	1920	1920

Note: Source: PSID. Standard errors in parentheses. Significance levels are denoted as follows: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors are heteroskedastic robust. The data uses sampling weights. Time, socio-demographic, cohort effects and other variables are included. Socio-demographic variables include age, sex and race of individuals. Panel (A) reports the effects of education on different quantiles of the distribution of wealth. Panel (B) reports effects of education on different quantiles of the distribution of wealth by age cohorts. Constant term is included but not reported for brevity.

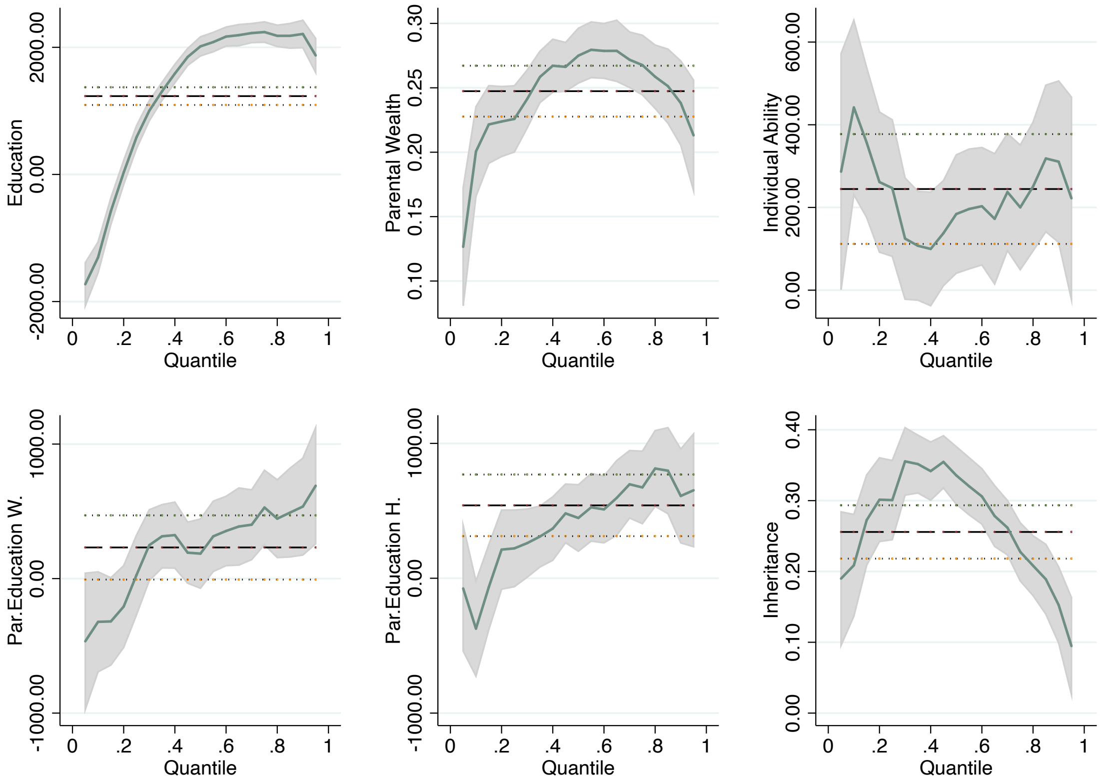


Figure A1: Education per Quantile of Wealth

Note: The graph shows the results of the quantile regression for some variables on household wealth, excluding home equity. Each panel has the estimates from the OLS regression with a black dashed line and confidence intervals. The solid lines are the estimates from the quantile regression with confidence intervals at 95%. The results are heteroscedasticity robust and sample-weighted. Source: Panel Study of Income Dynamics.

B life-cycle Model: Additional Information

B.1 life-cycle Model: Solution Method

As demonstrated by Carroll (2006), a method to facilitate the solution of these models is to rearrange the problem to reduce its state variables. Define the normalised variables $\mathbf{m}_t = m_t/h_t$, $\mathbf{c}_t = c_t/h_t$, $\mathbf{a}_t = a_t/h_t$, and $v_t(\mathbf{m}_t, \beta_t) = v(h_t, m_t, \beta_t)/h_t^{1-\gamma}$. Applying this transformation to the Bellman equation (which includes survival probability s_t and the bequest motive θ^e) yields:

$$v(\bar{z}_t) = \max_{c_t} \left\{ \frac{(\mathbf{c}_t h_t)^{1-\gamma}}{1-\gamma} + \beta_t \Big[s_t\, E_t v_{t+1}(\bar{z}_{t+1}) + (1-s_t)\, \theta^e(\mathbf{a}_t) \Big] \right\} \tag{B1a}$$

$$\frac{v(\bar{z}_t)}{h_t^{1-\gamma}} = \max_{c_t} \left\{ \frac{\mathbf{c}_t^{1-\gamma}}{1-\gamma} + \beta_t \Big[s_t\, E_t \frac{v_{t+1}(\bar{z}_{t+1})}{h_t^{1-\gamma}} + (1-s_t)\, \theta^e(\mathbf{a}_t) \Big] \right\} \tag{B1b}$$

$$v_t(\bar{\mathbf{z}}) = \max_{\mathbf{c}_t} \left\{ \frac{\mathbf{c}_t^{1-\gamma}}{1-\gamma} + \beta_t \Big[s_t\, E_t \Big[(G\psi_{t+1})^{1-\gamma} v_{t+1}(\bar{\mathbf{z}}_{t+1}) \Big] + (1-s_t)\, \theta^e(\mathbf{a}_t) \Big] \right\} \tag{B1c}$$

where $\bar{\mathbf{z}} = (\mathbf{m}, \beta)$ is the normalised state vector and $\theta^e(\mathbf{a}) = \theta_1^e(1 + \mathbf{a}/\theta_2)^{1-\gamma}$ is the education-specific bequest utility evaluated at end-of-period normalised assets. The normalised budget constraint is:

$$\mathbf{m}_{t+1} = \frac{R^e_{t+1}}{G\psi^e_{t+1}} \mathbf{a}_t + \xi^e_{t+1} \tag{B2}$$

where R^e_{t+1} is the education-specific gross return (Section 3). The superscript e on the shocks is a reminder that the discretisation nodes differ between college and non-college agents.

This normalisation reduces the problem to two state variables $(\mathbf{m}, \beta)$, making it feasible to solve by backward induction. The first-order condition with respect to $\mathbf{c}_t$ gives the Euler equation, which now contains two terms: a *survival* continuation and a *bequest* term:

$$u'(\mathbf{c}_t) = E_t \left[\beta_t R^e_{t+1}\, s_t\, \Big(G\psi^e_{t+1}\, \mathbf{c}^*_{t+1}(\mathbf{m}_{t+1}) \Big)^{-\rho} \right] + E_t[\beta_t]\, (1-s_t)\, (\theta^e)'(\mathbf{a}_t) \tag{B3}$$

The first term on the right is the standard discounted expected marginal utility of future consumption, scaled by the return and the survival probability. The second term is the expected marginal bequest utility, evaluated at current end-of-period savings $\mathbf{a}_t$ and weighted by the death probability $(1 - s_t)$. Crucially, the bequest term does *not* carry the return factor R^e_{t+1}: the bequest event occurs at the end of period t (after $\mathbf{a}_t$ has been chosen, before any new return is realised).

An alternative to value function iteration is the Endogenous Grid Method (EGM) of Carroll (2006). The convergence condition is equation (19). The shocks β_{t+1}, R^e_{t+1}, ψ^e_{t+1} and ξ^e_{t+1} are discretised by Gauss-Hermite quadrature with 8 nodes and weights $(\pi^i_\beta, \pi^i_R, \pi^i_\psi, \pi^i_\xi)$. Because income shock volatility differs by education type (Section 3),

separate quadrature grids are built for college and non-college agents, using $(\sigma_\psi^s, \sigma_\xi^s) = (0.08, 0.09)$ and $(\sigma_\psi^u, \sigma_\xi^u) = (0.13, 0.14)$ respectively. A mass point at $\xi = \mu$ with probability π is appended to each transitory shock grid to represent the unemployment state (equation 13).
The EGM algorithm is applied in two separate backward-induction passes, one per education type, so that each group's policy function is solved against the distributions and return mean it actually faces. A third pass using average shock parameters produces the aggregate policy function used in simulation diagnostics.
EGM Algorithm (for education type e):

1. Construct an asset grid $\mathbf{a} \in \Gamma_a \equiv \{a_1, a_2, \ldots, a_J\}$.
2. At the terminal period T: set $\mathbf{c}_T = \mathbf{m}_T$ (consume all wealth), so $\mathbf{a}_T = 0$. The terminal value function is $v_T(\mathbf{m}_T) = u(\mathbf{m}_T) + \theta^e(0)$.
3. For each period $t = T-1, \ldots, 1$, and for each $a_i \in \Gamma_a$:
   (a) Compute the expected marginal utility of future consumption (survival term) and the marginal bequest utility (death term) using equation (B3):

$$\mathbf{c}_i = \left\{ \underbrace{\mathrm{E}_t\left[\beta_t R_{t+1}^e\, s_t \left(G\psi_{t+1}^e\, \mathbf{c}_{t+1}^*\left(\frac{R_{t+1}^e}{G\psi_{t+1}^e} a_i + \xi_{t+1}^e\right)\right)^{-\rho}\right]}_{\text{survival continuation}} + \underbrace{\mathrm{E}[\beta_t]\,(1-s_t)\,(\theta^e)'(a_i)}_{\text{bequest term}} \right\}^{-\frac{1}{\rho}} \tag{B4}$$

   (b) Recover the endogenous cash-on-hand grid point:

$$\mathbf{m}_i = \mathbf{a}_i + \mathbf{c}_i \tag{B5}$$

4. Use the resulting pairs $\{(\mathbf{m}_i, \mathbf{c}_i)\}$ as the consumption policy function for period t, and repeat backward.

B.2 Estimation Probability of Attending College

The primary objective here is to explore the key factors influencing the decision to attend college. To achieve this, logistic regression was applied using the Panel Study of Income Dynamics data from 2019, offering a contemporary snapshot of how socioeconomic factors impact educational decisions during early adulthood.
The relationship between parental education, family wealth, and the probability of attending college is modeled as follows:

$$\text{logit}(P(\text{college})) = \beta_0 + \beta_1 \cdot \text{Par.Wealth} + \beta_2 \cdot \text{Par.Education} \tag{B6}$$

This equation encapsulates the log odds of college attendance as a function of parental wealth and education, suggesting that both factors may play a crucial role in shaping educational outcomes.

Table B1: Logistic Regression Results for Predicting College Attendance

	Coefficient	Std. Error
Par.Education	0.72003^{***}	(0.03299)
Par.Wealth	0.00000947^{***}	(0.00000230)
Constant	-1.5998^{***}	(0.09602)

Note: Significance levels are denoted as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Source: PSID, 2019.

The coefficients derived from the logistic regression model provide insights into the factors influencing college attendance. A positive coefficient for parental education suggests that an increase in the parents' educational attainment significantly raises the likelihood of their children attending college. Similarly, the coefficient for parental wealth indicates that even small increases in family wealth can enhance college attendance probabilities.

B.3 life-cycle Model: Inheritance's Fit

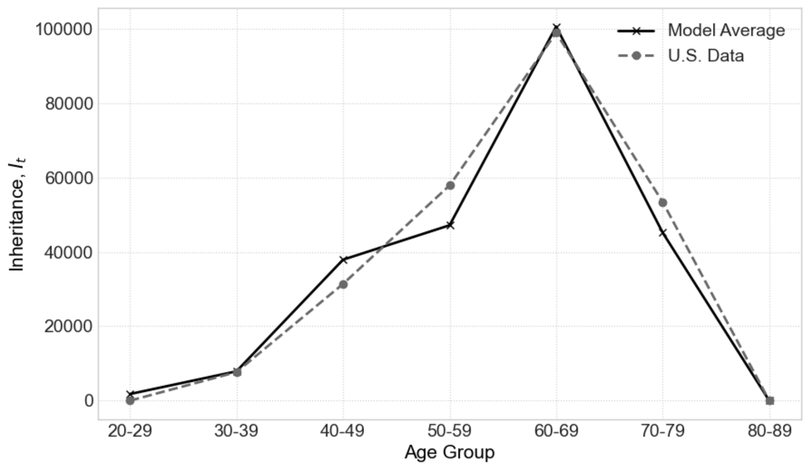


Figure B1: Average Inheritance by Age
Note: The figure compares the model's average inheritance received by individuals with the real data. Source: Panel Study of Income Dynamics, 2019.

B.4 Falsification Check: Parental Job Loss - Non-Tertiary

Table B2 reports a placebo-style reduced-form check that regresses wealth on parental job loss within the non-tertiary sample only. Because this restriction sharply reduces the usable sample and yields unstable cohort-specific estimates, the table reports the pooled specification only. The coefficient on parental job loss is negative but statistically insignificant, which is consistent with the interpretation that the main parental-job-loss IV results are operating through tertiary educational attainment rather than a direct wealth channel. This exercise should nevertheless be interpreted cautiously, since non-tertiary status is itself an outcome potentially affected by parental job loss.

Table B2: Placebo-Style Check: Parental Job Loss and Wealth in the Non-Tertiary Sample

	Avg
Parental Job Loss	-1022.36
	(1009.90)
Observations	6833

Note: The table reports a reduced-form random-effects regression of wealth on parental job loss in the non-tertiary sample only. Controls include survey-year effects, birth-cohort effects, age, sex, race, inheritance, family structure at age 16, parental wealth, IQ, and parental education. Standard errors clustered by individual are reported in parentheses. Because non-tertiary status is itself potentially affected by parental job loss, this is a suggestive falsification check rather than a formal test of the exclusion restriction.

B.5 Wealth Inequality by Age and Education

In the baseline calibration, college-educated individuals receive a permanent income premium $\Delta_s \approx \log(1.40)$, equivalent to a 40 percent level shift in permanent income relative to non-college individuals, consistent with lifetime earnings differentials documented in Card (1999). Income risk is education-specific: college agents face lower shock volatility $(\sigma_\psi^s, \sigma_\xi^s) = (0.08, 0.09)$ than non-college agents $(\sigma_\psi^u, \sigma_\xi^u) = (0.13, 0.14)$, following the estimates of Meghir and Pistaferri (2004). College agents also face a higher mean asset return ($\bar{u}_r^s = \log(1.12)$ versus $\bar{u}_r^u = \log(1.04)$), and a stronger bequest motive ($\theta_1^s > \theta_1^u$).
Table B3 reports wealth Gini coefficients generated by the model and observed in the PSID for 2019, by age group and education level. The model closely matches wealth inequality in mid and late adulthood, replicating a Gini coefficient of 0.79 for individuals aged 60–79, identical to the data. In early adulthood (20–39), the model generates lower wealth inequality than observed in the data, reflecting more limited dispersion in initial asset accumulation. Similarly, the model understates inequality among non-college individuals.
For college-educated individuals, the model generates a Gini coefficient of 0.75, somewhat below the 0.81 observed in the data, reflecting that the bequest motive calibration targets the overall wealth distribution rather than within-group inequality. The model also understates inequality among non-college households (0.64 vs. 0.80 in the data). These validation results are broadly satisfactory given that the policy experiments focus on the education-wealth gap across groups, rather than within-group inequality. The model captures the key cross-group distributional patterns most relevant to the analysis.

Table B3: Validation: Wealth Gini Coefficient

	U.S. Data	**Model**
Age		
Early Adulthood (20–39 y.o.)	0.82	0.67
Mid Adulthood (40-59 y.o.)	0.81	0.78
Late Adulthood (60-79 y.o.)	0.79	0.80
Education		
College	0.81	0.75
Non-College	0.80	0.64

Source for U.S. Data: Panel Study of Income Dynamics, 2019.

B.6 SMM Calibration: Target Fit

The bequest-motive parameters (θ_1^s, θ_1^u) are calibrated by minimising the distance between model-generated wealth distribution moments and SCF 2019 targets (Section 3.4). Table B4 reports the two targeted moments alongside the model's achieved values at the optimum. The baseline model closely matches both the overall wealth Gini and the top-decile share.

Table B4: SMM Calibration: Target Moments vs. Baseline Model

Moment	**Target (SCF 2019)**	**Baseline model**
Wealth Gini coefficient	0.82	0.83
Top-10% wealth share	76.7%	75.4%
Calibrated parameters: $\theta_1^s = 28.64$, $\theta_1^u = 15.91$, $\theta_2 = 11.37$ (common).		

Note: Moments are computed from the stationary wealth distribution of the baseline model (simulation=3, income and returns premiums active, $N = 1{,}000{,}000$ households per cohort). Target moments are drawn from the 2019 Survey of Consumer Finances. The common parameter $\theta_2 = 11.37$ is calibrated to the observed bequest-to-wealth ratio prior to the SMM step.

B.7 Simulation Outcomes: Wealth Distribution

Table B5 reports wealth distribution outcomes, namely overall Gini and top-share statistics, for each simulation discussed in Section 3.5. These complement the age-gradient regression results in Table 7 by showing the level and shape of the simulated wealth distribution.

Table B5: Simulation Outcomes: Wealth Distribution

	Gini	**Top 1%**	**Top 10%**	**Top 40%**	**Bot. 40%**
(a) Channel decomposition					
Returns Channel	0.81	32.1	72.9	94.4	2.0
Main Model	0.83	33.6	75.7	95.5	1.6
(b) Income risk mechanism					
Income Risk Equalization	0.85	35.1	77.9	96.2	1.3
(c) Policy counterfactuals					
Financial Democratization	0.76	24.2	63.6	91.9	2.7
Financial Education (+10% gap)	0.82	32.7	73.8	94.8	1.8
Education Expansion (+10% share)	0.83	33.0	75.1	95.5	1.5
U.S. Data (SCF 2019)	*0.82*	*37.4*	*76.7*	*96.4*	*0.2*

Note: Author's calculations. Columns report the Gini coefficient and the share of total net worth held by the top 1%, top 10%, top 40%, and bottom 40% of households (N = 1,000,000 per cohort). Returns Channel activates only the returns premium (no income premium). Income Risk Equalization sets non-college income-shock volatility to college levels ($\sigma^u_\psi, \sigma^u_\xi$ reduced to $\sigma^s_\psi, \sigma^s_\xi$). Financial Democratization gives all agents the college return process. Financial Education closes 10% of the returns gap for non-college agents. Education Expansion increases the college share by 10% (34% $\rightarrow$ 37.4%). U.S. Data from the 2019 Survey of Consumer Finances.

B.8 Model Regression Results: Full Tables with Std. Errors

Table B6 reproduces the education-wealth age-gradient results from Table 7 with standard errors and significance levels included. All simulations use N = 1,000,000 agents per cohort; standard errors are accordingly small by construction and statistical significance is not in question. The table is included for replication transparency and to allow comparison with the empirical benchmarks, which carry non-trivial standard errors.

Table B6: Education Coefficient on IHS(Net Worth) by Age Cohort

	$\hat{\beta}_{30s}$	$\hat{\beta}_{40s}$	$\hat{\beta}_{50s}$	$\hat{\beta}_{60s}$	**Grad.**	**% Data**
(a) Channel Decomposition						
Income Channel	-0.3195^{***}	-0.2006^{***}	0.0420^{***}	0.3061^{***}	-0.13	-137%
	(0.0034)	(0.0042)	(0.0047)	(0.0053)		
Returns Channel	1.2216^{***}	1.5418^{***}	1.8139^{***}	2.1826^{***}	1.48	59%
	(0.0032)	(0.0042)	(0.0051)	(0.0059)		
Main Model	1.2583^{***}	1.5582^{***}	1.8229^{***}	2.1926^{***}	1.45	54%
	(0.0035)	(0.0045)	(0.0054)	(0.0063)		
CSL IV (data)	$20{,}842.7^{+}$	$28{,}197.4^{***}$	$38{,}114.4^{***}$	$67{,}838.9^{**}$	1.83	100%
	(10,863.9)	(5,829.8)	(7,650.7)	(24,456.2)		
(b) Income Risk Mechanism						
Income Risk Equalization	1.6507^{***}	1.9614^{***}	2.0903^{***}	2.2979^{***}	1.27	32%
	(0.0032)	(0.0042)	(0.0050)	(0.0058)		
CSL IV (data)	$20{,}842.7^{+}$	$28{,}197.4^{***}$	$38{,}114.4^{***}$	$67{,}838.9^{**}$	1.83	100%
	(10,863.9)	(5,829.8)	(7,650.7)	(24,456.2)		
(c) Policy Counterfactuals						
Financial Democratization	-0.0380^{***}	0.0241^{***}	0.1053^{***}	0.2283^{***}	-2.77	-455%
	(0.0034)	(0.0046)	(0.0056)	(0.0066)		
Financial Education	1.0595^{***}	1.3338^{***}	1.6022^{***}	1.9694^{***}	1.51	62%
	(0.0035)	(0.0045)	(0.0054)	(0.0063)		
Education Expansion	1.2618^{***}	1.5617^{***}	1.8251^{***}	2.1912^{***}	1.45	54%
	(0.0034)	(0.0044)	(0.0053)	(0.0062)		
CSL IV (data)	$20{,}842.7^{+}$	$28{,}197.4^{***}$	$38{,}114.4^{***}$	$67{,}838.9^{**}$	1.83	100%
	(10,863.9)	(5,829.8)	(7,650.7)	(24,456.2)		

Note: Dependent variable is IHS(net worth). Each row reports the college coefficient $\hat{\beta}_{as}$ from a model-simulated OLS regression by age cohort, with standard errors in parentheses (N = 1,000,000 agents per cohort). Gradient = $\hat{\beta}_{50s}/\hat{\beta}_{30s}$; % Data = (Gradient − 1)/(Data gradient − 1) × 100. Empirical benchmark (CSL IV): compulsory-schooling-law IV on PSID (dollar units; not comparable in scale to IHS model coefficients). Significance: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.